

Visualization Tool For Electric Vehicle Charge And Range Analysis



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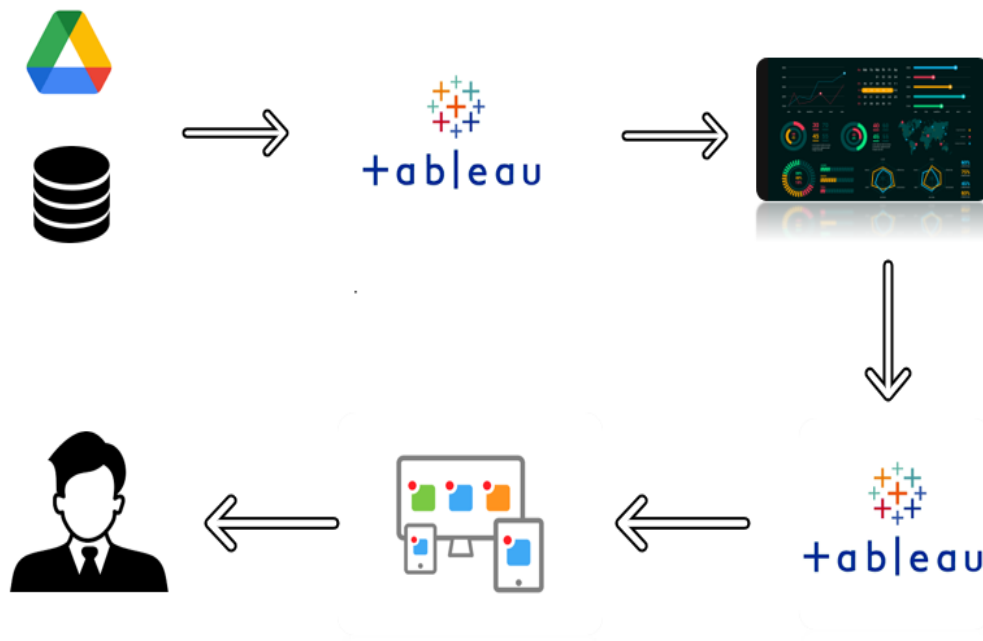
Vehicle Charge and Range Visualization Tool for Electric Analysis

The Electric Vehicle (EV) is not new, but it has been receiving significantly more attention in recent years. Advances in both EV analytics and battery technologies have led to increased automotive market share. However, this growth is not attributed to hardware alone. The modern vehicle marries electrical storage and propulsion systems with electronic sensors, controls, and actuators, integrated closely with software, secure data transfer, and data analysis, to form a comprehensive transportation solution. Advances in all these areas have contributed to the overall rise of EV's, but the common thread that runs through all these elements is data analytics.

Problem Statement:

Analysis different data from Multiple sources for Electric cars in India and Globally. We have 4 Different datasets we need to the data and create Dashboard and story that can represent the data and show the Visuals for the data.

Technical Architecture:



Technical architecture is the structural design and blueprint of an IT system, defining how software, hardware, networks, and data components interact to meet business requirements. It ensures systems are secure, scalable, and reliable by defining technical specifications, patterns, and infrastructure. It serves as a guide for development and, when documented, facilitates system maintenance and evolution.

Key Aspects of Technical Architecture:

- **Components & Interaction:** Defines the relationships between software (applications, databases) and hardware (servers, networks).
- **Non-Functional Requirements:** Focuses on scalability, reliability, security, and performance rather than just functionality.
- **Design Decisions:** Involves choosing technology stacks, platforms, and architectural patterns (e.g., microservices, cloud-native).
- **Documentation:** Utilizes diagrams to visualize system structure, data flow, and component interactions.
- **Scope:** While enterprise architecture covers the entire organization, technical architecture often focuses on specific application or system-level implementation.

Common Technical Architecture Types:

- **Application Architecture:** Focuses on the structure of a specific software application.
- **Infrastructure Architecture:** Covers the underlying hardware, networks, and cloud services.
- **Data Architecture:** Defines how data is stored, managed, and processed.
- **Security Architecture:** Ensures the system is designed with protective measures.

Benefits:

- **Risk Mitigation:** Identifies potential issues early, preventing costly rework.
- **Consistency:** Provides a, unified framework for development teams.
- **Efficiency:** Ensures optimal use of existing, infrastructure and technology.



Pre requisites

For Completing this project these are some of the prerequisites needed

- A system with a minimum 4GB RAM and 128GB Hard Disk
- Good Internet Connection
- Google Drive / Any of the Database Server with Management Studio
- <https://youtu.be/2c2fUOgZMmY>
- **SQL Server Management Studio:**
- <https://youtu.be/kGdTg-vGs-E>
- **Tableau Desktop:**
- <https://youtu.be/b3pWYyrHQo8>
- Tableau Public Account: <https://public.tableau.com/app/discover>
- Html, CSS or Bootstrap

Prior Knowledge

To Complete this project, one must understand the below concepts and able to work with the tools

Data Visualization:

- <https://youtu.be/5gpnZvMSTZs>
- <https://youtu.be/JG8GRIMjp3c>
- **Chart Types:**
- <https://youtu.be/csXmVBw8cdo>
- **Tableau:**
- <https://youtu.be/aHaOIvR00So>
- **Business Intelligence:**
- <https://youtu.be/Hg8zBJ1DhLQ>

Project Objectives

By the end of this project, you will:

- Able to Connect Tableau with different data sources
- Know fundamental concepts and techniques used for Data Visualization.
- Gain a broad understanding about data and different types of charts.
- Have knowledge on developing Visualizations, Dashboards and Story.
- Able to Integrate the developed dashboard and story with the web application

Project Flow

To accomplish this, we have to complete all the activities listed below,

- Data collection
 - Collect the dataset or create the dataset
- Database /Spreadsheet Connection
 - Understand the dataset
 - Import Dataset into the database
 - Connect Tableau Desktop to Database server.
- Visualizing and analysis data
- Understand the Data and the Business Questions
- Based on the Business questions develop the different visualizations
- Dashboard

Develop the Dashboard

- Story

Develop the Storyboard

- Publishing to the Tableau Public & Web Application Integration
 - Developed Visualizations, Dashboard and story will be published to Tableau Public Account.
 - Once it is published, we will get the shareable links
 - Develop a web application using HTML, CSS or Using Bootstrap
 - Integrate the Visualizations, Dashboard and Story with the Web Application

Data Collection

Data collection is the process of gathering and measuring information on variables of interest, in an established systematic fashion that enables one to answer stated research questions, test hypotheses, and evaluate outcomes and generate insights from the data.

Downloading the dataset

Please use the link to download the dataset:

<https://drive.google.com/drive/folders/1Rkzdks6Us1Uq2SRB4nxMAb83jN5bpHll>

Working with Dataset

Here we will start working and understanding the dataset

Understand the data

Data contains all the meta information regarding the columns described in the CSV files. we have provided 4 CSV files:

1. EVIndia
2. Electric_vehicle_charging_station_list
3. ElectricCarData_Clean
4. Cheapestelectriccars-EVDatabase

Column Description for EVIndia:

- Car - Car Brand name and model
- Style Range - Style range of car
- Transmission- Transmission type
- VehicleType – Type of vehicle
- PriceRange(Lakhs) - Price Range in Lakhs
- Capacity - Capacity of car
- BootSpace – Boot space of the car

- BaseModel – Base model name
- TopModel – Top model name

Column Description for Electric vehicle charging station list:

- region: This column represents the region of the charging station.
- address: This column represents the address of the charging station.
- aux address: This column represents the auxiliary address of the charging station.
- latitude: This column represents the latitude of the charging station.
- longitude: This column represents the longitude of the charging station
- type: This column represents the type of the charging station.
- power: This column represents the power of the charging station.
- service: This column represents the type of service at the charging station.

Column Description for ElectricCarData Clean:

- Brand
- Model
- AccelSec
- TopSpeed_KmH
- Range_Km
- Efficiency_WhKm
- FastCharge_KmH
- RapidCharge
- PowerTrain
- PlugType
- BodyStyle

- Segment
- Seats
- PriceEuro

Column Description for Cheapestelectriccars-EVDatabase:

- Name
- Subtitle
- Acceleration
- TopSpeed
- Range
- Efficiency
- FastChargeSpeed
- Drive
- NumberofSeats
- PriceinGermany
- PriceinUK

Import Dataset into Database

Explanation video link: [Database creation](#)

Explanation video link: [Basic SQL Operations](#)

Connect Tableau Desktop to Database server

Explanation video link: [Link](#)

Data Preparation

Preparing the data for visualization involves cleaning the data to remove irrelevant or missing data, transforming the data into a format that can be easily visualized, exploring the data to identify patterns and trends, filtering the data to focus on specific subsets of data, preparing the data for visualization software, and ensuring the data is accurate and complete. This process helps to make the data easily understandable and ready for creating visualizations to gain insights into the performance and efficiency.

Link: https://drive.google.com/file/d/1lAMzG-Cut2uKqrYv7Z1gHtBJZ7XtM1YT/view?usp=share_link

Data Visualization

Data visualization is the process of creating graphical representations of data in order to help people understand and explore the information. Data visualisation aims to make complex data sets more accessible, intuitive, and easier to interpret. By using visual elements such as charts, graphs, and maps, data visualizations can help people quickly identify patterns, trends, and outliers in the data.

Charging Stations by region and type in India

Explanation, video

link: https://drive.google.com/file/d/1QuWuNC6S0wd04n8kCH_Z7Bo9PnZZvaEK/view?usp=share_link

EV Charging stations map of India

Explanation video link: https://drive.google.com/file/d/1JgMlZ7pKEIoYvOe5X0v59kmVTFZ-nSQV/view?usp=share_link

Different EV cars in India

Explanation video

link: https://drive.google.com/file/d/1PvVm2oqr8j1ERO2IuaNUcsROI3QY2ffC/view?usp=share_link

Top speed for different Brands

Explanation video

link: https://drive.google.com/file/d/1VXW8C9b4ycVHaEXnAJobgB_N_miSl9V/view?usp=share_link

Price for different cars in India

Explanationvideo

link: https://drive.google.com/file/d/115HwkYlphwVgQGu8LHxBmuJJKH9dyfhp/view?usp=share_link

Top 10 most efficient EV Brands

Explanationvideo

link: https://drive.google.com/file/d/1EhhapHlPsEqEE21e_2sLHQtr0VDXt8d/view?usp=share_link

Brands according to Bodystyle

Explanationvideo,link: https://drive.google.com/file/d/16KiZpHumlDdb88ggZ8ZzvrOqpAJw8gg/view?usp=share_link

Brand filtered by PowerTrain type

Explanationvideo

link: https://drive.google.com/file/d/1Gnrj7h6A7il6p26TLTNfRVQTYnREoQwn/view?usp=share_link

No of models by each brand

Explanationvideolink: https://drive.google.com/file/d/1-FZplgBanEa6HHql9M-Iu6YchyQszskC/view?usp=share_link

Summary card for Different brands of EV Cars globally

Explanationvideolink: https://drive.google.com/file/d/1CaL3ZLOypWlH77xmabJg4K8fKNlxaVtH/view?usp=share_link

Summary card for Different brands of EV Cars in India

Explanationvideolink: https://drive.google.com/file/d/1C61Jxi4jOCdoVbrRNWluDSJse6dguMPS/view?usp=share_link

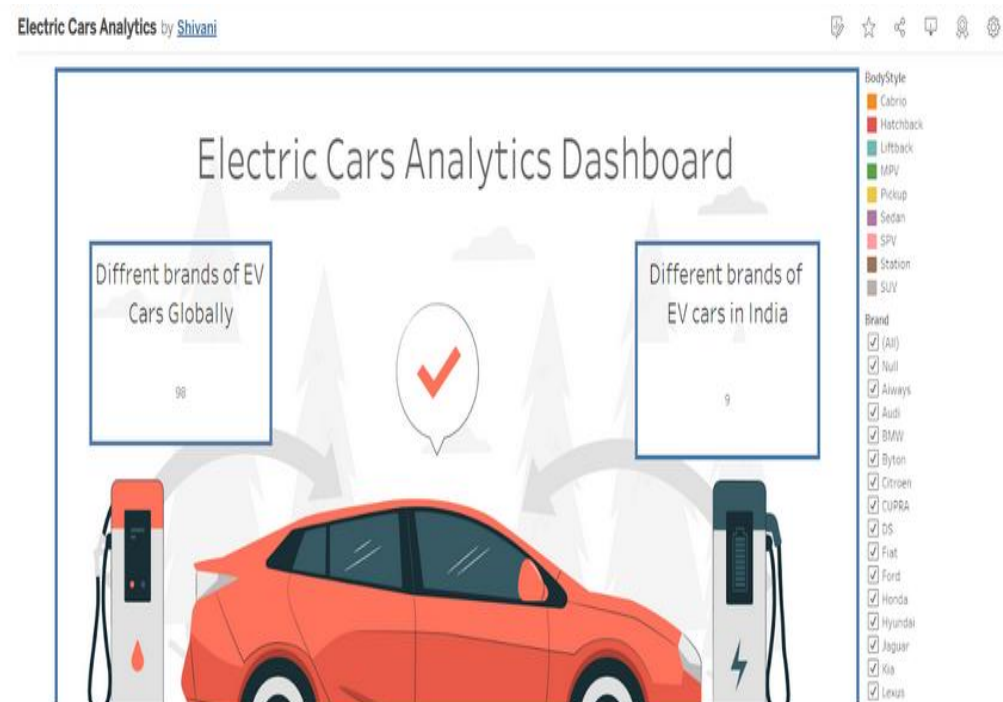
Dashboard

A dashboard is a graphical user interface (GUI) that displays information and data in an organized, easy-to-read format. Dashboards are often used to provide real-time monitoring and analysis of data, and are typically designed for a specific purpose or use case. Dashboards can be used in a variety of settings, such as business, finance, manufacturing, healthcare, and many other industries. They can be used to track key performance indicators (KPIs), monitor performance metrics, and display data in the form of charts, graphs, and tables.

Creating the Dashboard

Once you have created views on different sheets in Tableau, you can pull them into a dashboard.

Explanation video link: [Link](#)



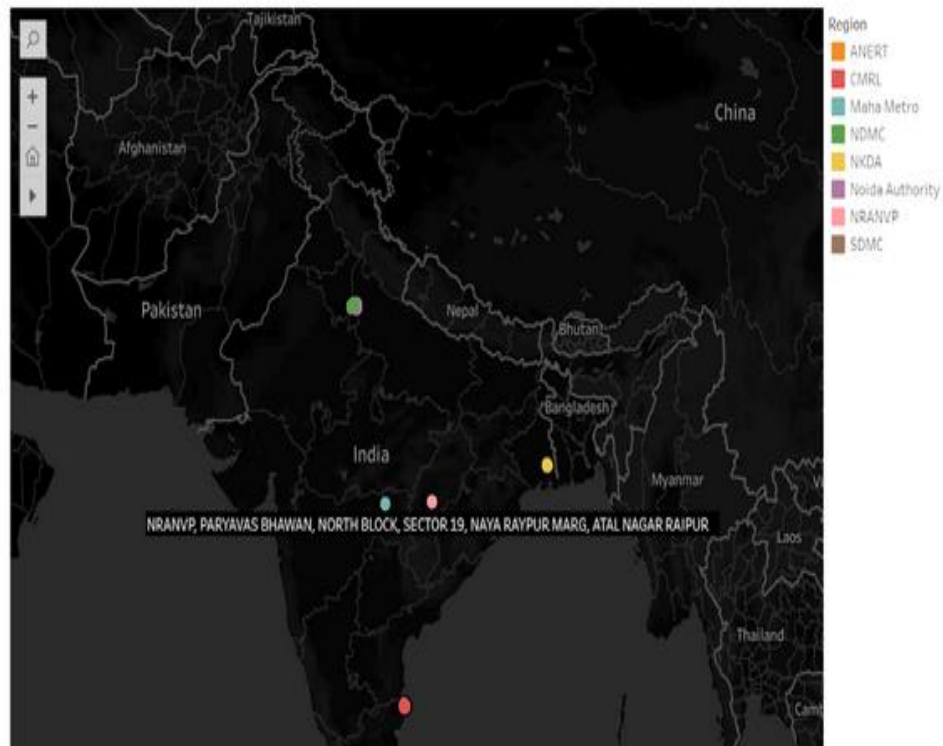
Story

A data story is a way of presenting data and analysis in a narrative format, with the goal of making the information more engaging and easier to understand. A data story typically includes a clear introduction that sets the stage and explains the context for the data, a body that presents the data and analysis in a logical and systematic way, and a conclusion that summarizes the key findings and highlights their implications. Data stories can be told using a variety of mediums, such as reports, presentations, interactive visualizations, and videos.

Creating the story board

Explanation video link: [Link](#)

Story of Electric Cars In India

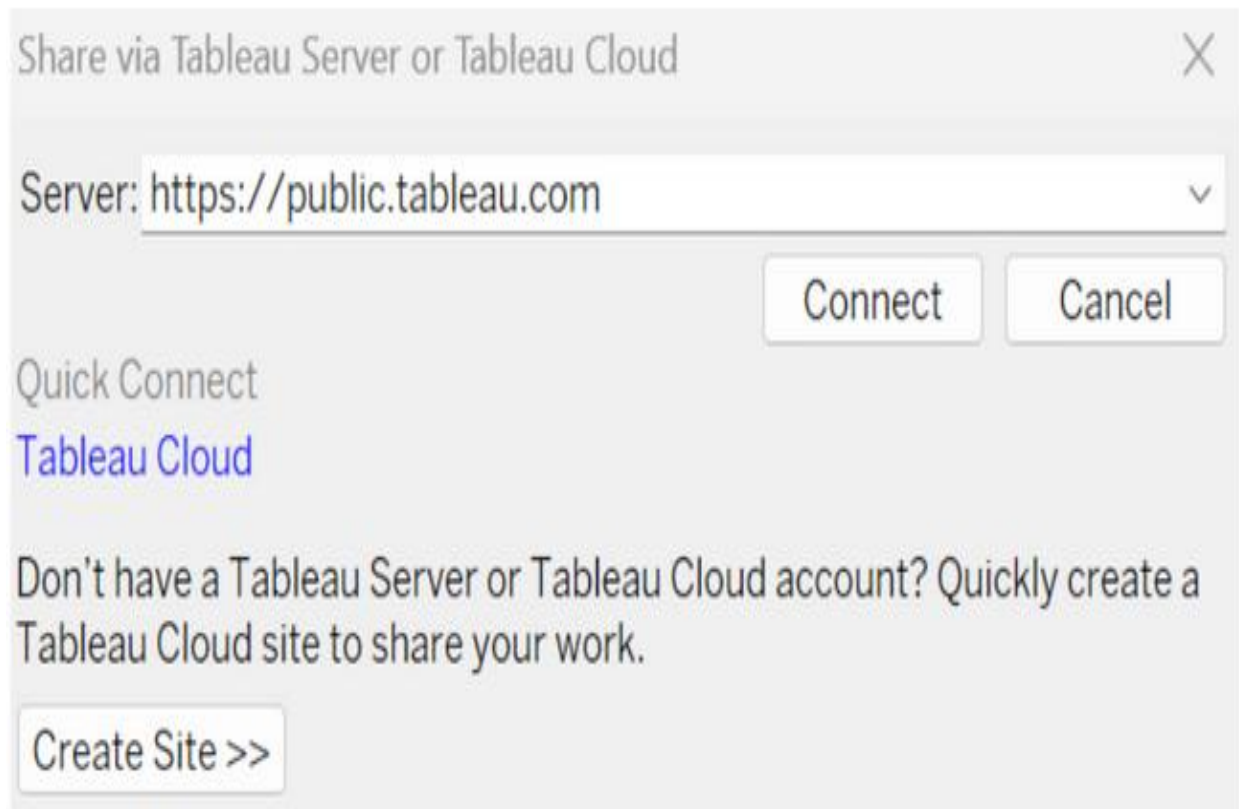


Publishing and Web integration

Publishing helps us to track and monitor key performance metrics, and to communicate results and progress. help a publisher stay informed, make better decisions, and communicate their performance to others.

Publishing dashboard and reports to tableau public

Step 1: Go to Dashboard/story, click on the share button on the top ribbon



Give the server address of your tableau public account and click on connect.

Explanation Video:- [Link](#)

step 2: Once you click on connect it will ask you for tableau public user name and password



Email

Password

Sign In

 This site is SSL encrypted

[Forgot your password?](#)

[Don't have a profile yet?](#)

[Create one now for free](#)

Once you login into your tableau public using the credentials, the particular visualization will be published into tableau public

Note: While publish the visualization to the public, respective sheet will get published where you click on share option.

Embed Dashboard & Story with Web Bootstrap

Explanation video link: [Link](#)

Electric Vehicles

All-electric vehicles—also referred to as battery electric vehicles (BEVs)—plug-in hybrid electric vehicles (PHEVs), and hybrid electric vehicles (HEVs) all use electricity, which improves vehicle efficiency. In colloquial references, these three vehicle types are sometimes called electric cars, electric-drive vehicles, electric vehicles, or simply EVs even though some of these vehicles still use liquid fuels in conjunction with electricity. BEVs and PHEVs are also referred to more specifically as plug-in electric vehicles (PEVs). [View a short introductory video](#) for more information on electric vehicle technology.

[All-Electric Vehicles](#)



All-electric vehicles, also called battery electric vehicles, have a battery that is charged by plugging the vehicle in to charging equipment. These vehicles always operate in all-electric mode and have typical driving ranges from 150 to 400 miles.

Plug-In Hybrid Electric Vehicles



PHEVs are powered by an internal combustion engine and an electric motor that uses energy stored in a battery. PHEVs can operate in all-electric (or charge-depleting) mode. To enable operation in all-electric mode, PHEVs require a larger battery, which can be plugged in

Hybrid Electric Vehicles



HEVs are powered by an internal combustion engine and one or more electric motors that use energy stored in a battery. The vehicle is fueled with gasoline to operate the internal combustion engine, and the battery is charged through regenerative braking, not by plugging in.

A Vehicle Charge and Range Visualization Tool is a data-driven, interactive dashboard—often built with Python (Pandas, Matplotlib) or Tableau—designed to analyze, predict, and map electric vehicle (EV) charging behavior, battery efficiency, and driving range. It enables users to visualize charging station availability, model range based on energy consumption, and optimize charging schedules, reducing range anxiety.

Key components of these visualization tools include:

- **Data Sources & Processing:** The tool utilizes datasets containing EV models, battery capacity (kWh), efficiency (

Wh/kmWh/km

Wh/km

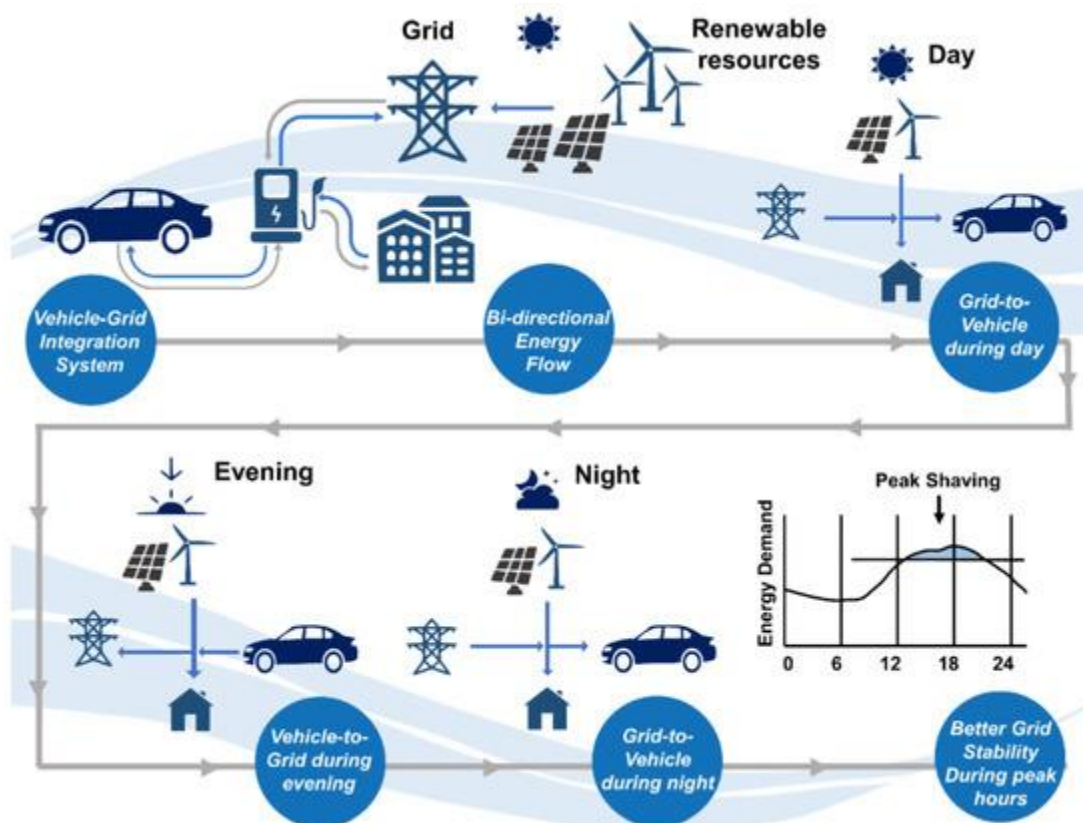
), and charging station locations (latitude/longitude). Data is cleaned to remove missing values and processed to analyze performance trends.

- **Visualization & Dashboarding:** [LightningChart Python](#) or Tableau are used to create interactive maps, bar charts for charging station distribution, and scatter plots for battery efficiency.
- **Range Analysis & Prediction:** Machine learning models (e.g., Random Forest or SVM) are applied to predict real-world range based on factors like speed, charging session length, and vehicle type.
- **Key Metrics Displayed:**
 - **Range**

Range (km)=Battery Capacity (Wh)÷Consumption (Wh/km) Range (km) equals Battery Capacity (Wh) divided by Consumption (Wh/km)

Range (km)=Battery Capacity (Wh)÷Consumption (Wh/km).

- **Charging Infrastructure:** Visualization of public chargers, including power type (fast vs. slow) and location.
- **Charging Behavior:** Analysis of, for example, the 80/20 rule to maximize battery health.



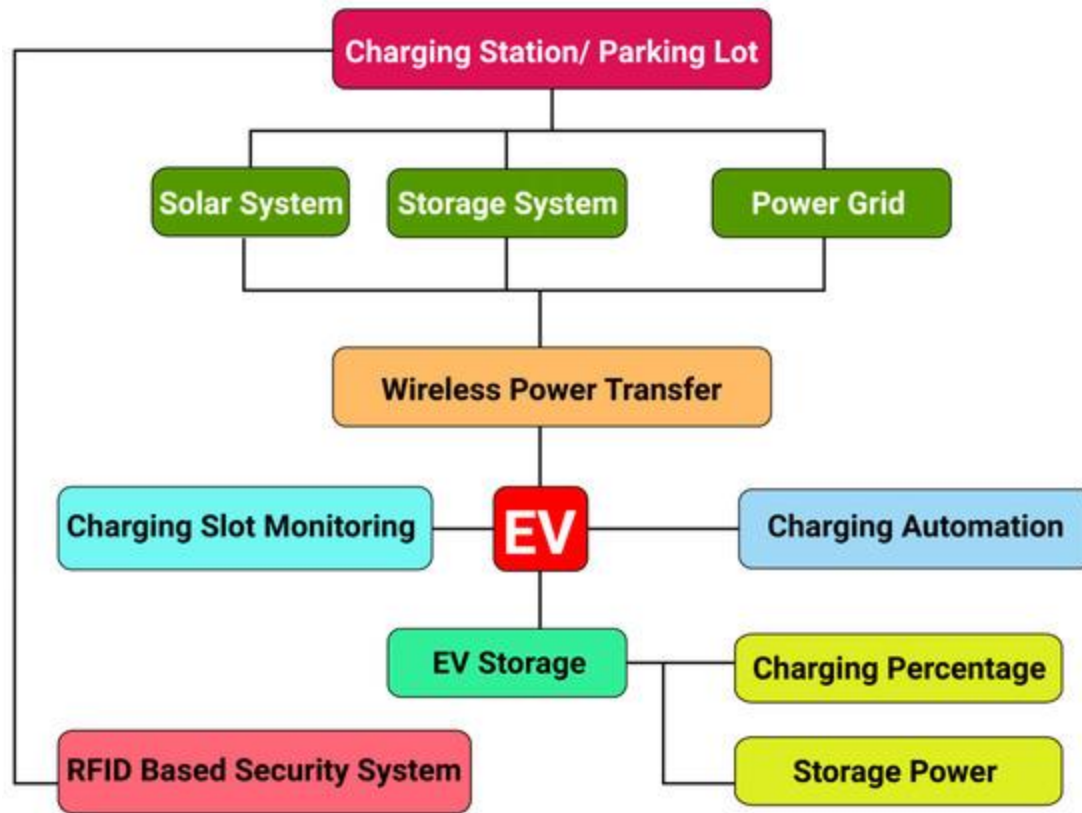
Various **vehicle charge and range visualization tools** are available, ranging from consumer-facing mobile apps to professional data analytics platforms and open-source software libraries for electric vehicle (EV) analysis and planning.

Consumer and Planning Tools

- **PlugShare:** One of the most widely used community-powered apps for locating charging stations, it allows users to filter by connector type, charging speed, and real-time availability, which helps in visualizing potential range and charging stops during trips.
- **ChargePoint:** This app offers access to a major charging network and provides tools to find stations along a route, essential for planning longer journeys and visualizing charging options.
- **Electrify America:** A strong option for those prioritizing fast charging, the app helps easily locate DC fast chargers, often situated near major travel corridors, facilitating quick top-ups.
- **Zapmap:** Features an EV route planner that provides peace of mind on long journeys by showing a wide choice of charging options and real-time availability.
- **Mapbox for EV:** Offers services for carmakers and developers, providing live range visualization and optimized EV routing based on continuously updated data, including battery status, ambient temperature, and real-time traffic/weather conditions.

Data Analysis and Simulation Platforms

- **EVPRE (Electric Vehicle Range Prediction Estimator):** An open-source, Python-based software tool that generates energy-optimal routes and range predictions for any EV. It accounts for specific vehicle information, elevation changes (using Google Maps Elevation API), weather (Open Weather API), and traffic data (TomTom Traffic API) to provide highly accurate, customizable range estimations.
- **emobpy:** An open-source, Python-based tool that creates profiles of battery-electric vehicles based on empirical mobility statistics and customizable assumptions, primarily for generating data time series for research and policy making.
- **IEA Electric Vehicle Charging and Grid Integration Tool:** Designed for planners and policymakers, this tool quantifies and visualizes EV charging demand profiles for different vehicle classes and use cases to estimate the impact on the electricity grid.



- **Intangles EV Monitoring Solutions:** Utilizes Ambient Cognitive AI to provide precise distance-to-empty (DTE) and state-of-charge (SoC) predictions for fleet operators, incorporating factors like driving behavior and weather.
- **GT-SUITE:** A multi-physics simulation software used by manufacturers to predict and certify EV range under various standard and real-world driving cycles, leveraging detailed vehicle and battery models.

Technical Components for Developers

- **LightningChart Python:** A high-performance data visualization library used in Python to build advanced, real-time analytics dashboards for electric vehicle data, capable of handling large datasets for metrics like charging frequency and energy consumption.
- **Tableau:** A popular commercial data visualization tool used in various EV analysis projects to create interactive dashboards and maps of charging station locations, types, and usage trends from gathered datasets.

- **Python with Pandas and NumPy:** These libraries form the backbone of many custom analysis projects, used for collecting, processing, and analyzing raw EV data before visualization with libraries like Plotly or Dash.

Electric vehicle range prediction estimator (EVPRE)

Original software publication. Electric vehicle range prediction estimator (EVPRE) ... Highlights EVPRE software provides ener...

- **Find EV Charging Stations Near Me (2026): 3 Best Apps - Coltura**
1 Feb 2026 — These apps work like an EV charger locator — helping you filter by speed, connector type, pricing, and recent station activity. * ...
- **Visualization Tool for Electric Vehicle Charge and Range**
Handbook SmartBridge Educational Services Pvt. Ltd. * Visualization Tool for Electric Vehicle Charge and Range. Analysis. Project ...

Specialized EV Analysis & Simulation Tools

- **EVI-Pro (Electric Vehicle Infrastructure-Projection Tool):** Developed by the National Renewable Energy Laboratory (NREL), this tool is designed for infrastructure planning and analysis.
 - It estimates the required quantity and type of charging infrastructure for a designated area based on vehicle technology, user demographics, and travel patterns.
- **ADVISOR (Advanced Vehicle Simulator):** An older, open-source tool, often implemented in MATLAB/Simulink, that allows for detailed vehicle performance and energy consumption analysis through simulations.
 - Users can simulate different drive cycles and analyze how factors like battery capacity and regenerative braking affect the vehicle's range.
- **emobpy:** An open-source, Python-based tool that creates profiles of battery-electric vehicles based on empirical mobility statistics.
 - It generates time series data for vehicle location, driving electricity consumption, and actual charging electricity demand.

- **EV-ChART (Electric Vehicle Charging Analytics and Reporting Tool):** A web-based data portal and analytics platform used by federal funding recipients to submit and analyze charging station data for infrastructure development. Access is limited to authorized users.

Data Visualization and Custom Dashboards

Researchers and developers often create custom visualization tools using programming languages and libraries to analyze specific datasets.

Scribd

- **Python with Libraries (Pandas, NumPy, LightningChart, Plotly, Dash):** This is a robust platform for data analytics and visualization.
 - **LightningChart Python** is noted for handling large datasets and creating high-speed, high-quality scientific and industrial visualizations.
 - **Plotly** and **Dash** are used for creating interactive, web-based dashboards that can visualize charging station locations on maps and display trends in station density and usage.
- **Tableau:** This commercial business intelligence and data visualization tool is frequently used to create interactive dashboards for analyzing EV charging and range data.
 - Dashboards can visualize data on charging station types, regional availability, and other performance metrics.
- **E-DRIVE (Evaluation and Development of Regional Infrastructure for Vehicle Electrification):** A mapping application built with Tableau software that allows users to rank and display priority locations for EV fast-charging infrastructure installation.

Key Data Insights Provided by These Tools

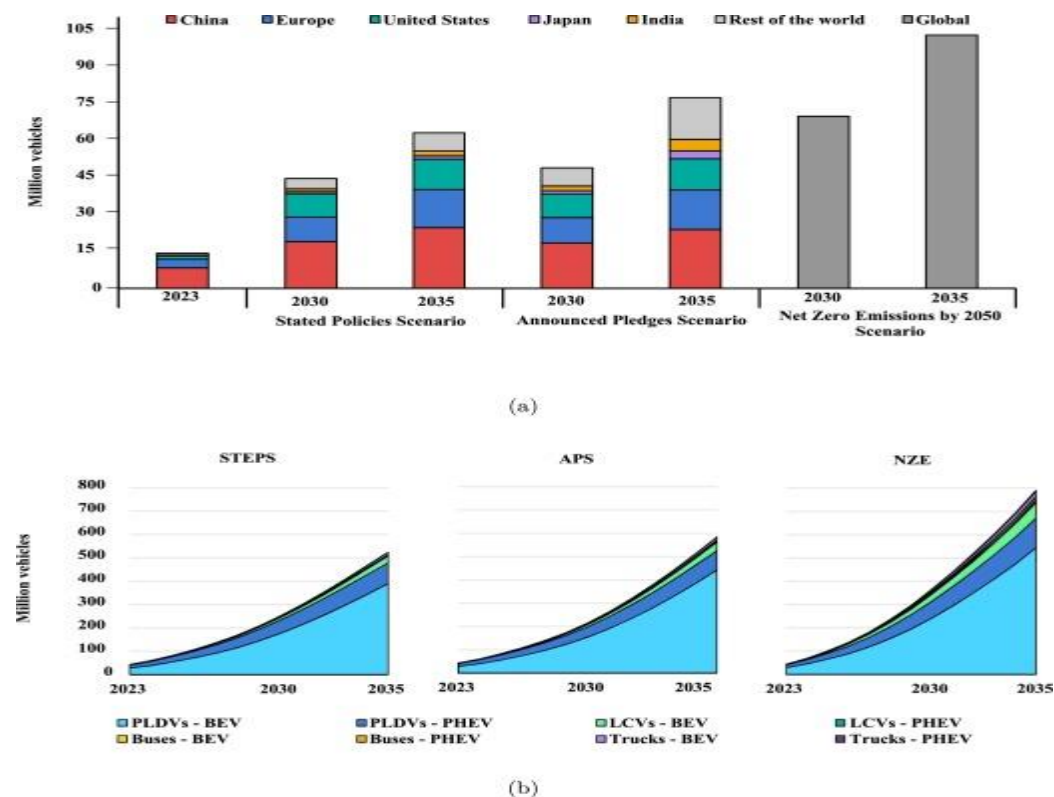
These tools and platforms help to visualize and understand several critical aspects of EV use and infrastructure:

- **Geospatial Visualization:** Mapping charging station locations, identifying underserved areas, and planning infrastructure development.
- **Charging Behavior Analysis:** Understanding user patterns, such as peak charging times, preferred power types (AC/DC), and session duration.
- **Range Prediction:** Using machine learning to predict real-world energy consumption and remaining range based on factors like weather, traffic conditions, and battery health.
- **Energy Management:** Optimizing resource allocation for charging point operators and forecasting charging demand to prevent grid overload.

- EV Charge and Range Analysis Project - Electric Vehicle - Scribd
7 Dec 2025 — EV Charge and Range Analysis Project. The 'Electric Vehicle Charge and Range Analysis' project focuses on analyzing the availabili...
- Electric Vehicle Data Analytics - LightningChart
24 Apr 2025 — Electric vehicle data analytics app. ... With the increasing penetration of EVs, effective data analytics is crucial to optimize c...

LightningChart

- EV Charging Infrastructure Mapping Tools
EV Charging Infrastructure Mapping Tools * Electric vehicles (EVs) have enormous potential to reduce harmful air pollution and gre...



Wireless charging infrastructure (WCI) for electric vehicles (EVs) uses

[magnetic resonance coupling](#) to transfer power from ground-embedded pads to vehicle-mounted receivers, supporting both static (parking) and dynamic (moving) charging to enhance

convenience. Visualization tools analyze these systems to optimize battery performance, range, and station location.

Scribd +4

Key Aspects of Wireless Charging & Visualization

- **Technological Approach:** Inductive Power Transfer (IPT) employs high-frequency magnetic fields to transfer energy, reducing the need for cable handling.
- **Charging Modes:**
 - **Static Wireless Charging (SWC):** Vehicles charge while parked over a pad, often coupled with automatic parking systems.
 - **Dynamic Wireless Charging (DWC):** Power is transferred to the vehicle while it moves over electrified road tracks.
- **Infrastructure Components:** Systems consist of primary (ground) and secondary (vehicle) coils, featuring IoT-based sensors for detecting vehicle presence and controlling charge flow.
- **Visualization & Analysis Tools:** Software solutions track and analyze real-time data, including battery state of charge (SoC), remaining range, and charging efficiency.
- **Market Growth:** The wireless EV charging market is expected to grow from

\$0.09\$ 0.09

\$0.09

billion in 2025 to

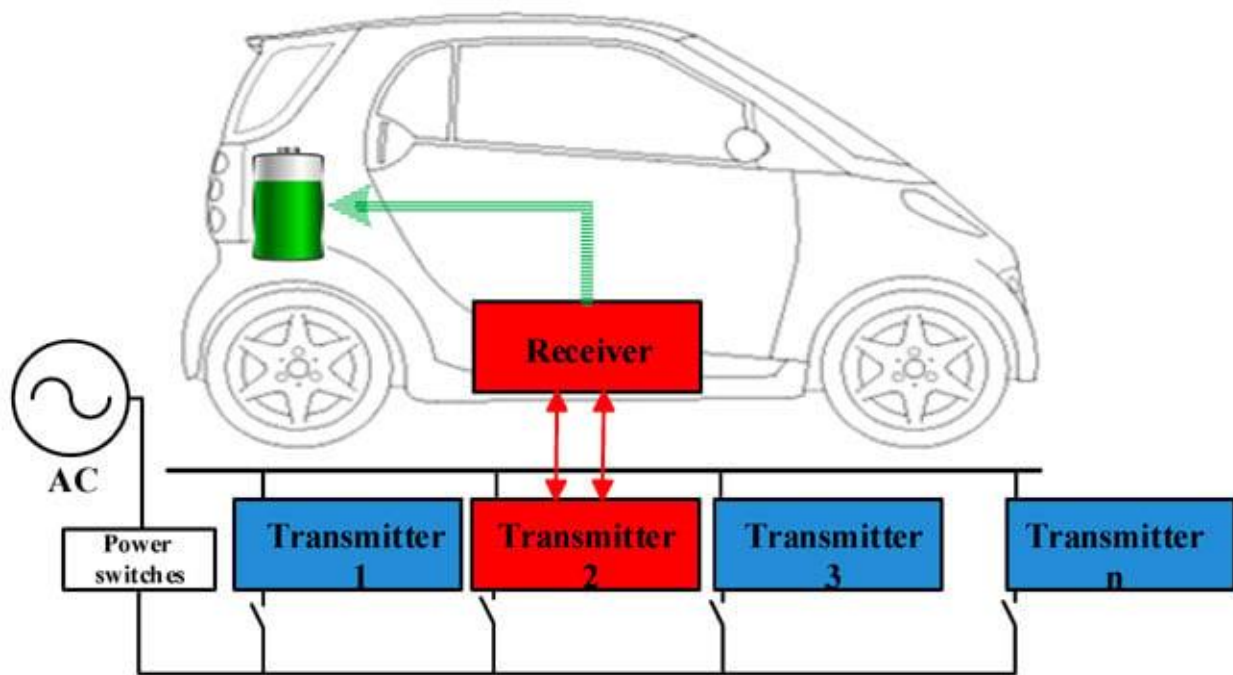
\$1.12\$ 1.12

\$1.12

billion by 2032, driven by the demand for seamless charging experiences.

Scribd +6

- **Enhanced User Experience:** Eliminates the need for manual cable connection, enabling cleaner, safer, and automated charging at home or in public.
- **Extended Range:** DWC reduces the need for large, heavy batteries by allowing smaller batteries to be charged while driving.
- **Strategic Planning:** Data visualization allows for analyzing the best locations for new charging stations to optimize network infrastructure.



The cheapest EV in India is currently the Vayve Mobility Eva (around ₹3.25 Lakh) or PMV EaS E (around ₹4.79 Lakh) for ultra-compact, low-speed options, while for mainstream, road-legal cars, the [MG Comet EV](#) (starting ~₹6.31 Lakh ex-showroom) and the [Tata Tiago EV](#) (starting ~₹7.99 Lakh ex-showroom) are the top contenders for budget-friendly electric mobility.

Top Budget-Friendly EVs (Ex-Showroom):

1. Vayve Mobility Eva / PMV EaS E:

These are micro-EVs, often categorized as quadricycles, ideal for very prices starting under ₹5 Lakh.

2. [MG Comet EV](#):

A popular choice for city driving due to its compact size and futuristic design, starting around ₹6.31 Lakh, offering a 230 km range.

3. [Tata Tiago EV](#):

A more practical, 4-door hatchback with better range options (up to 293 km on the larger battery) and a smooth ride, starting from approximately ₹7.99 Lakh.

4. [Tata Punch EV](#):

A family-friendly mini-SUV option, often available under ₹10 Lakh for its base variants, offering space and good safety.

Key Considerations:

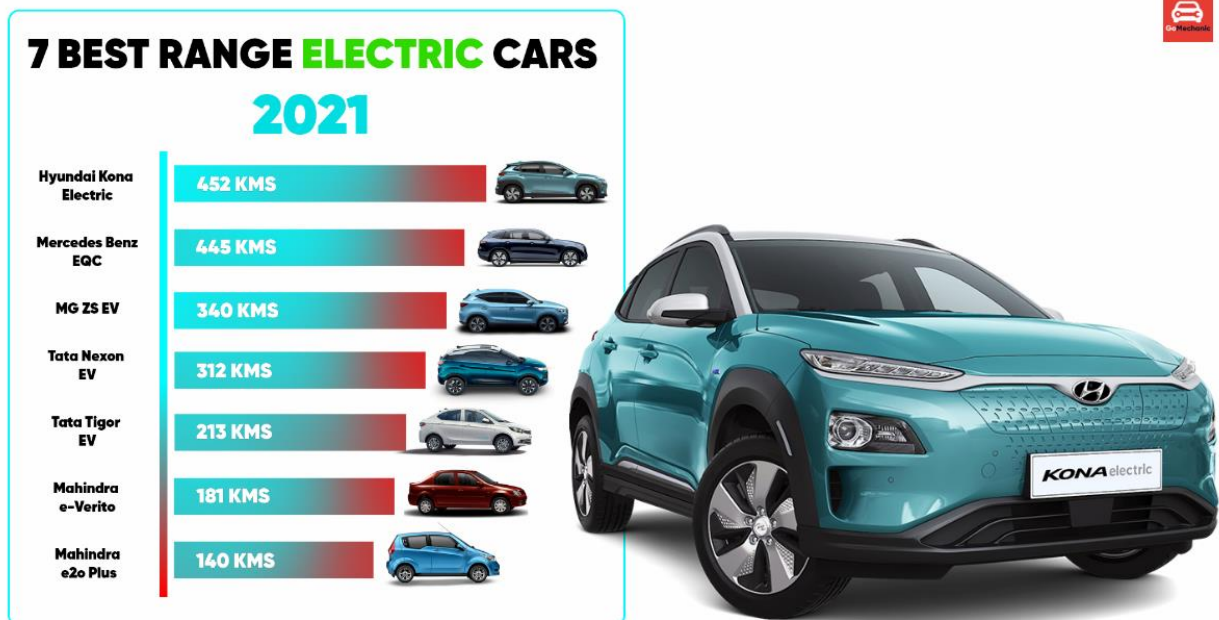
- **Price Range:**

Expect entry-level EVs to start from ₹5 Lakh to ₹10 Lakh (ex-showroom), with higher variants and features increasing the cost.

-

- **On-Road vs. Ex-Showroom:**

Prices listed are usually ex-showroom; on-road prices will be higher due to RTO and insurance costs.



The Mercedes-Benz EQS leads with the highest claimed range in India, offering over 800 km, while the Kia EV6 and BYD Seal follow as top luxury, long-range options. For more accessible options, the Mahindra BE 6 and XEV 9e provide over 500-600 km ranges, with the Tata Nexon EV and MG ZS EV serving as popular, reliable choices.

Top Long-Range Electric Cars in India (2025-2026)

- **Mercedes-Benz EQS:**

Claimed range of

≈ is approximately equal to

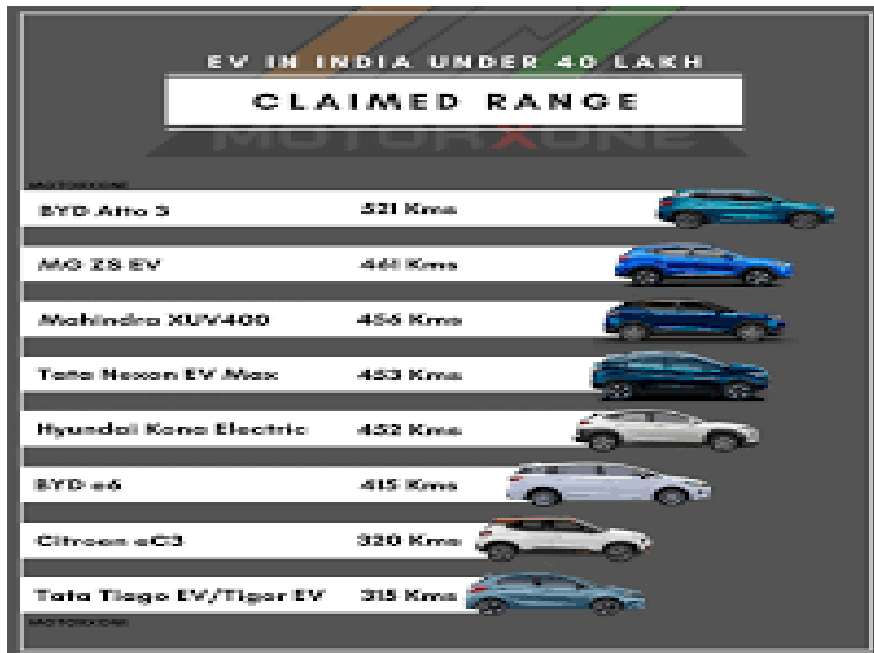
≈

813–857 km

- **Kia EV6:** Claimed range of 582–708 km.
- **BYD Seal:** Claimed range of 510–650 km.
- **Mahindra BE 6:** Claimed range of 557–683 km.
- **Mahindra XEV 9e:** Claimed range of 542–656 km.
- **Tata Harrier EV:** Claimed range of 538–627 km.
- **BMW i7:** Claimed range of 560–625 km.
- **MG ZS EV:** Claimed range up to 550 km (Ultra).
- **BYD Atto 3:** Claimed range up to 521 km.
- **Tata Curvv EV:** Claimed range up to 502 km.

Key Considerations

- **Real-World Range:** While claimed (ARAI/NEDC) figures are high, actual range depends on driving conditions and traffic, often lower than advertised.
- **Charging Speed:** High-range vehicles like the Kia EV6 and Hyundai Ioniq 5 support fast charging, significantly reducing downtime.
- **Segment Leaders:** For premium, the BMW i7 and Mercedes-Benz EQS lead, while the Mahindra XEV 9e and Tata Harrier EV offer competitive range in the SUV segment.



Specialized Modeling & Simulation Software

- **MATLAB:** The industry standard for 1D dynamic modeling. It is used to simulate battery management systems (BMS), regenerative braking algorithms, and powertrain architectures.
- **EVI-X Modeling Suite:** A specialized toolkit from the National Renewable Energy Laboratory (NREL) that includes **EVI-Pro** for projecting charging demand and **EVI-RoadTrip** for long-distance travel analysis.
- **EV-Insights:** An open-source framework designed to standardize data ingestion and analysis for large public EV charging datasets.

Public Mapping & Real-Time Driver Apps

- **Plug Share:** A community-driven tool essential for visualizing real-world charger reliability, connector types, and real-time availability.
- **Zap-Map:** A leading UK-based app for advanced route planning and live status updates of the charging network.
- **Tata Power EZ CHARGE:** A prominent app in India that provides real-time session insights, including State of Charge percentages and units consumed.

Analyzing and visualizing electric vehicle (EV) charge and range data requires tools that can handle high-frequency sensor data, geospatial mapping, and complex battery performance metrics

Electric vehicles (EVs) have moved from a niche technology to a central pillar of global transportation, driven by environmental goals and massive improvements in battery technology

Main Types of Electric Vehicles

The "EV" label covers several distinct technologies based on how they use electricity and fuel:

- **Battery Electric Vehicles (BEV):** Fully electric cars powered only by a battery. They have no gas engine and produce zero tailpipe emissions.
- **Plug-in Hybrid Electric Vehicles (PHEV):** Feature both a battery/motor and a gasoline engine. They can run on electricity for short trips (typically 20–50 miles) and switch to gas for longer journeys.
- **Hybrid Electric Vehicles (HEV):** Use a small battery to assist a gas engine. They cannot be plugged in; the battery recharges through "regenerative braking" while driving.
- **Fuel Cell Electric Vehicles (FCEV):** Use hydrogen gas to generate electricity onboard. They only emit water vapor but currently have limited refueling infrastructure.

Core Benefits & Performance

- **Lower Running Costs:** Electricity is generally cheaper and less price-volatile than gasoline. Maintenance is also simpler since BEVs have about 90% fewer moving parts—no oil changes, spark plugs, or exhaust systems to fix.
- **Performance:** Electric motors provide "instant torque," meaning they accelerate much faster and more smoothly than gas engines.
- **Efficiency:** EVs convert over **77%** of their electrical energy into power at the wheels, whereas gas cars typically convert only about **12%–30%** of fuel energy into motion.

Range and Charging Realities

While "range anxiety" remains a concern, modern BEVs typically offer **150 to 400 miles** per charge. Actual range is affected by:

Department of Energy (.gov) +1

- **Speed:** Driving faster (especially over 100 km/h) increases air resistance and drains the battery exponentially faster.

- **Weather:** Extreme cold can temporarily reduce range by **20%–30%** as the battery uses energy to heat itself.
- **Charging Speed:** Home charging (Level 2) usually takes **4–10 hours** for a full charge, while public "DC Fast Chargers" can reach 80% in **20–40 minutes**.

MG Motor India +5

Market Trends (2025–2026)

- **Global Sales:** Over **20% of new cars** sold globally in 2024 were electric. This is expected to surpass **25%** in 2025.
- **Leaderboard:** **China** is currently the world's largest EV market, followed by Europe and the United States. Leading manufacturers include **Tesla, BYD,** and the **Volkswagen Group**.
- **Affordability:** Average battery prices have dropped significantly (from over \$600/kWh in 2010 to under \$140/kWh recently), helping push many models toward price parity with gas cars.

1,Executive Summary

The primary challenge of Electric Vehicle (EV) adoption remains the optimization of **Range (distance per charge)** and **Charging Efficiency (speed and health)**. This report analyzes the variables affecting these metrics, including battery chemistry, thermal management, aerodynamics, and user behavior.

2. Fundamental Battery Metrics

At the heart of the analysis is the Lithium-Ion battery. Two key metrics define the "fuel tank" of an EV:

- **State of Charge (SoC):** The percentage of the usable battery capacity remaining.
- **State of Health (SoH):** The ratio of current maximum capacity compared to the original capacity.
- **Energy Density:** Measured in

Wh/kgcapW h / k g

Wh/kg

. Higher energy density directly correlates to more range without adding significant weight.

3. The Physics of Range: Factors of Depletion

Range isn't just about battery size; it's about energy loss. The power required to move an EV is calculated by:

$$P = (F_{roll} + F_{air} + F_{grad} + F_{accel}) \times v$$

P equals open paren cap F sub r o l l end-sub plus cap F sub a i r end-sub plus cap F sub g r a d end-sub plus cap F sub a c c e l end-sub close paren cross v

- **Aerodynamic Drag:** Since EVs are quiet, wind resistance is the primary energy drain at highway speeds.
- **Rolling Resistance:** The friction between tires and the road.
- **Auxiliary Loads:** HVAC systems (heating/cooling) can reduce range by up to 25% in extreme climates.

4. Charging Levels and Infrastructure

Charging is classified into three levels based on voltage and current

Level	Voltage	Type	Typical Charge Time
Level 1	120V AC	Household Outlet	40-50 hours (full)
Level 2	240V AC	Home/Work Wallbox	4-10 hours (full)
Level 3	400V-800V DC	DC Fast Charger	20-40 mins (to 80%)

5. The Charging Curve: Why 80% is the Limit

Charging speed is not linear. As a battery fills, internal resistance increases.

- **CC (Constant Current) Phase:** The charger delivers max power until the battery reaches roughly 80%.
- **CV (Constant Voltage) Phase:** The charger tapers off the power to prevent overheating and cell damage.

6. Thermal Management Systems (TMS)

Batteries are like humans; they perform best between **20°C and 35°C (68°F - 95°F)**.

- **Cold Weather:** Slows chemical reactions, increasing internal resistance.
- **Hot Weather:** Accelerates degradation.
- **Active Cooling:** Modern EVs use liquid cooling loops to maintain the "Goldilocks zone."

7. Regenerative Braking Analysis

Unlike gas cars, EVs can recover energy. When the driver lifts off the accelerator, the motor acts as a generator, converting kinetic energy back into chemical energy in the battery. This can improve city-driving range by **10-15%**.

8.Degradation and Lifecycle Analysis

Battery health declines over time due to:

1. **Cycle Aging:** Each full charge/discharge cycle causes microscopic wear.
2. **Calendar Aging:** Natural degradation over time, regardless of use.
- **Depth of Discharge (DoD):** Consistently discharging to 0% is significantly more harmful than keeping the battery between 20% and 80%.

9. Comparative Data: BEV vs. ICE Efficiency

This section compares the **Well-to-Wheel** efficiency. EVs typically achieve over **85% efficiency**, whereas Internal Combustion Engines (ICE) waste roughly **70%** of their fuel energy as heat.

10. Future Trends: Solid-State and V2G

- **Solid-State Batteries:** Promise to double energy density and eliminate fire risks.
 - **Vehicle-to-Grid (V2G):** Allows EVs to sell power back to the grid during peak demand, turning cars into mobile power plants
-

1.Battery Chemistry & Raw Materials

The battery pack is the most material-intensive component of an EV.

- **Cathode Materials:** Most modern EVs use **Lithium Nickel Manganese Cobalt (NMC)** or **Lithium Iron Phosphate (LFP)**. NMC batteries provide higher energy density, while LFP is prized for safety and lower cost.
- **Anode Materials:** **Graphite** is the standard material due to its cycle stability. Ongoing research explores **Silicon** or **Lithium Metal** to increase energy density.
- **Other Components:** **Lithium** ions facilitate energy flow, while **Copper** and **Aluminum** act as current collectors.

2. Rare Earth Elements (REE) in Electric Motors

EV motors rely on powerful permanent magnets to achieve high efficiency.

- **Neodymium, Dysprosium, and Praseodymium:** These REEs are essential for **Brushless Permanent Magnet (BPM)** motors. They allow motors to be compact while delivering high torque.
- **Copper Windings:** High-purity copper is used in the stator to conduct electricity and create the electromagnetic field necessary for propulsion.

3. Lightweight Structural Materials

Reducing vehicle weight is critical to extending range.

- **Aluminum Alloys:** Aluminum is roughly **50% lighter than steel** and is used for body panels, chassis, and battery enclosures due to its high strength-to-weight ratio.
- **Magnesium Alloys:** The lightest structural metal available, magnesium is used for seat frames, steering wheels, and cross-car beams.
- **Carbon Fiber Reinforced Polymers (CFRP):** Extremely strong and light, CFRP is used in high-performance EVs for structural reinforcement and battery protection, though it remains expensive.

4. Advanced Power Electronics Materials

Power electronics manage the flow of electricity between the battery and the motor.

- **Wide-Bandgap Semiconductors:** Materials like **Silicon Carbide (SiC)** and **Gallium Nitride (GaN)** are replacing traditional silicon in inverters. They operate at higher temperatures and voltages with significantly lower heat loss, improving overall vehicle efficiency.

5. Specialized Polymers and Composites

EVs use a variety of plastics for insulation, safety, and weight reduction.

- **Thermoplastic Resins:** Polypropylene (PP) and ABS are used for interior trims and under-the-hood components.
- **Battery Separators:** Polyethylene or polypropylene films act as membranes between the anode and cathode, preventing short circuits while allowing ion flow.

6. Environmental Impact of Material Extraction

The "hidden cost" of EVs lies in the extraction of raw materials.

- **Lithium Mining:** Requires vast amounts of water (approx. **2.2 million liters per ton**), often in arid regions like South America's "Lithium Triangle".
- **Cobalt and Nickel:** Mining these can lead to toxic chemical leaks, soil degradation, and ethical concerns regarding labor practices in regions like the DRC.

7. Sustainable and Bio-Based Alternatives

Manufacturers are pivoting toward eco-friendly materials to lower production emissions.

- **Bio-Composites:** Fibers from **hemp, flax, and kenaf** are used in door panels and seat structures as renewable alternatives to glass fiber.
- **Vegan Interiors:** Materials like **Piñatex** (from pineapple leaves) or cork are replacing animal leather and wood trims.

8. The Critical Role of Copper

EVs require significantly more copper than gas-powered cars—roughly **183 pounds per EV** compared to 47 pounds for internal combustion vehicles.

- **Applications:** Beyond the motor, copper is the "circulatory system" of the EV, used in miles of internal wiring, charging cables, and high-voltage busbars.

9. Material Recycling and Circular Economy

As the first generation of EVs retires, recycling is becoming a primary source of materials.

- **Urban Mining:** Specialized processes can recover up to **95% of lithium, cobalt, and nickel** from spent batteries.
- **Closed-Loop Systems:** Companies like Tesla and Volkswagen are developing facilities to reintroduce recycled metals directly back into the battery production line.

10. Future Materials: The Next Generation

Emerging technologies aim to solve current material limitations.

- **Solid-State Electrolytes:** Replacing liquid electrolytes with solid ceramics or polymers could eliminate the need for certain flammable materials and allow for faster charging.
- **Graphene:** Being researched for use in batteries to dramatically improve conductivity and charging speeds, though it is not yet in mass production.

A wide range of electric vehicle (EV) models and types are available globally and in India, spanning from affordable city cars to high-performance luxury SUVs and sports cars. These models primarily fall into the Battery Electric Vehicle (BEV) or Plug-in Hybrid Electric Vehicle (PHEV) categories.

Currently Available and Popular EV Models

Popular EV models range across various segments, from budget-friendly options ideal for urban commutes to premium models offering extended range and luxury features.

- **Mass Market / Compact:**
 - **MG Comet EV:** A compact electric car known for its small size and suitability for urban commutes, offering a range of around 230 km.

- **Tata Punch EV:** A popular choice in India for its balance of features, range, and SUV styling, with an estimated range between 315 km and 421 km depending on the variant.
- **Tata EV:** A well-established electric SUV with a range of up to 489 km, noted for its practicality and performance.
- **Tesla Model Y:** The top-selling BEV globally in 2025, praised for its efficiency, technology, and performance.

Premium / Luxury:

- **Hyundai 5:** Features a distinctive design, spacious interior, and a claimed range of 631 km.
- **Kia EV6:** A sleek crossover built on an advanced 800V architecture enabling ultra-rapid charging, with a high claimed range of up to 663 km.
- **BMW :** A luxury SUV offering comfort, a high-end cabin, and a range of around 575 km.
- **BYD Seal:** A sedan known for its excellent range (up to 650 km) and fast-charging capabilities.

Upcoming EV Models (2026)

The upcoming models for 2026 focus on expanding the variety of segments and introducing more affordable options.

- **Suzuki :** Expected to launch in early 2026, this SUV will come with multiple battery options and a claimed range of up to 543 km.
- **Tata Sierra EV:** A highly anticipated electric version of the classic Sierra, expected to launch later in 2026, targeting the premium SUV segment.
- **Audi Q6 :** A new luxury electric SUV expected to offer a range of around 625 km and advanced features.
- **Tesla Model 3:** An updated version of the popular sedan is expected to launch in India towards the end of 2026.

Types of EVs

- **Battery Electric Vehicles (BEVs):** These run purely on electricity stored in a battery pack and have no tailpipe emissions. The models listed above (Tesla Model Y, Tata Nexon EV, etc.) are all BEVs.
- **Plug-in Hybrid Electric Vehicles (PHEVs):** These combine an electric motor/battery with a gasoline engine. The battery can be charged from an external source, allowing for a limited all-electric range before the engine takes over. Examples include the Audi Q5 TFSIe PHEV and the Volvo XC90 T8 PHEV.
- **Hybrid Electric Vehicles (HEVs):** These also use both an engine and motor but cannot be plugged in. The battery is charged by the engine and regenerative braking.

- **Fuel Cell Electric Vehicles (FCEVs):** These generate electricity from hydrogen and emit only water vapor.

Electric bikes, encompassing both electric scooters and motorcycles, offer an eco-friendly and cost-effective transportation alternative to traditional petrol-powered two-wheelers, with models ranging from budget-friendly commuters to high-performance sports bikes.

Types of Electric Two-Wheelers

- **Electric Cycles (E-Cycles):** These resemble traditional bicycles but include an electric motor and battery to provide pedal assist or full throttle power. They are generally speed-limited (around 25 km/h) and often do not require registration or a license in India.
- **Electric Scooters/Mopeds:** These are designed for urban commuting, typically featuring a step-through frame, ample under-seat storage, and a focus on practicality and comfort. They generally have lower top speeds than motorcycles.
- **Electric Motorcycles:** Built for higher performance, these models prioritize speed, range, and a traditional motorcycle feel. They feature stronger motors, longer ranges, and advanced features like liquid cooling and connectivity.

Popular Models and Key Features (2026)

The electric two-wheeler market in India is competitive, with manufacturers focusing on range, technology, and price to attract consumers.

- **Ather450X:** A tech-loaded, sporty electric scooter featuring a dashboard, navigation, and over-the-air (OTA) updates. Reviewers praise its performance and smart features.
- **Revolt RV400:** An electric motorcycle known for its practical range and the unique option of a portable, swappable battery for convenient charging.
- **TVS e:** A balanced electric scooter designed for eco-conscious city riders, offering a reliable range and connected tech via a TFT display.
- **Bajaj :** Combines a classic, retro design with modern electric technology, featuring a sturdy metal body and a reliable performance for urban commutes.

A high-performance electric sports bike that offers an exhilarating ride experience with a top speed of over 150 km/h and a substantial range.

Key Benefits

- **Environmentally Friendly:** Electric bikes produce zero tailpipe emissions, contributing to reduced air and noise pollution in urban environments.
- **Low Operating Costs:** Electricity is significantly cheaper than petrol, resulting in substantial savings on running and maintenance costs (e.g., no oil changes or spark plugs).
- **Instant Torque:** Electric motors deliver immediate, smooth acceleration (instant torque from 0 RPM) for a responsive and often exhilarating ride, ideal for city traffic.
- **Government Support:** The Indian government has encouraged adoption through initiatives and subsidies like the FAME II scheme, making EVs more affordable for consumers.

Here is the distribution of electric vehicle (EV) market data for 2025 and 2026, visualizing global and domestic trends through key metrics.

Global EV Market Share (2025)

BYD has solidified its position as the global leader, commanding nearly **20%** of the world market share. Tesla remains a dominant force but has seen its share decline as competition from Chinese manufacturers intensifies.

- **BYD:** 19.3% – 19.9% share
- **Tesla:** 7.7% – 8.0% share
- **Geely:** 5.6% – 10.0% share
- **Wuling:** 3.8% share
- **Volkswagen:** 2.8% share

India EV Market Share by Manufacturer (Jan 2026)

In India, **Tata Motors** continues to lead the four-wheeler segment, though its absolute market share has faced pressure from a surging **MG Motor** and **Mahindra**.

- **Tata Motors:** ~53% (FY25) / 40% (Jan 2026)
- **MG Motor:** ~28% (FY25) / 28.3% (Jan 2026)
- **Mahindra:** ~18.5% (Jan 2026)
- **VinFast:** ~2.4% (Emerging player as of Jan

Ola Electric's Sales Zoom 64%

Company	December	January	MoM Change (%)
 GREAVES Empowering Lives	2,840	3,411	20.11
 ATER	10,508	12,101	15.16
 BAJAJ	18,369	20,176	9.84
 HEROELECTRIC The smart move	58	78	34.48
 Hero	1,022	1,493	46.09
 LECTRIX	562	247	-56.05
 oben electric	60	179	198.33
 OLA ELECTRIC	13,794	22,656	64.25
 PURE	1,088	1,580	45.22
 REVOLT	997	1,022	2.51
 ULTRAVIOLETTE	62	76	22.58
 River	195	588	201.54
 TVS	17,333	22,080	27.39
Total	66,888	85,687	28.11

Note: Data as on December 31, 2025
Source: Vahan Dashboard

Inc42

EV Sales Distribution by Vehicle Type (India FY25)

Two-wheelers remain the primary driver of EV adoption in India, making up more than half of all electric units sold.

- **Electric 2-Wheelers:** 59.4%
- **E-Rickshaws (3W):** 23.3%
- **Electric 4-Wheelers (Cars):** 5.7%
- **High-Speed 3-Wheelers:** 6.6%

Powertrain Trends: BEV vs. PHEV

While Battery Electric Vehicles (BEV) dominate the market, Plug-in Hybrids (PHEV) are seeing significant growth in specific regions like China.

- **Global BEV Share:** Roughly **64.2%** of the total EV market.
- **Global PHEV Share:** Roughly **35.8%** of the total EV market.
- **UK Projection (2026):** BEVs are expected to capture **28.2%** of all new car registrations.

India's electric vehicle (EV) sales reached a record **2.27 million units in 2025**, a **16.3% increase** over the 1.95 million units sold in 2024. This growth was driven primarily by electric two-wheelers (E2Ws) and a significant surge in electric passenger vehicles (E4Ws).

India EV Sales by Category (2025)

The E2W segment continues to dominate the market by volume, while the E4W segment saw the highest year-on-year growth.

India EV Sales by Category (Calendar Year 2025)				
E-2 Wheelers				
E-3 Wheelers (L5/Rickshaw)				
E-4 Wheelers (PV)				
0500,000	1,000,000	1279951	8138351	76560

Monthly EV Passenger Car Sales (Jan 2026)

In **January 2026**, electric passenger vehicle sales hit **18,042 units**, marking a **51% year-on-year increase** and the third-best monthly performance ever.

Autocar India +1									
Monthly Electric Passenger Car Sales (Aug 2025 - Jan 2026)									
05,000	10,000	15,000	Aug 2025	Sep 2025	Oct 2025	Nov 2025	Dec 2025	Jan	
2026	18563	16368	19195	17873	15626	18042			

Market Share by Manufacturer (Jan 2026)

Electric Passenger Vehicles (E4W)

Tata Motors remains the market leader, reconfirming its dominance in January 2026 with its highest-ever monthly sales.

- **Tata Motors:** 7,852 units (**43.5% share**)
- **JSW MG Motor:** 4,606 units (**25.5% share**)
- **Mahindra:** 3,589 units (**20.0% share**)
- **VinFast:** 432 units

Electric Two-Wheelers (E2W)

A major shift occurred in January 2026 as **TVS Motor** surpassed all competitors to claim absolute leadership, while previous leader **Ola Electric** saw a significant decline.

- **TVS Motor:** 34,440 units (**28.1% share**)
- **Bajaj Auto:** 25,520 units (**20.8% share**)
- **Ather Energy:** 21,923 units (**17.9% share**)
- **Ola Electric:** 7,511 units (**6.1% share**)

Global electric vehicle (EV) sales are bifurcated into two distinct trajectories: cars dominate in developed markets like Europe and North America, while electric two-wheelers (E2Ws) lead adoption in emerging economies across Asia and Africa.

Worldwide EV Sales Distribution (2025–2026)

1. Electric Cars (Passenger Vehicles)

Global electric car sales reached **20.7 million units** in 2025, a 20% increase from 2024.

Benchmark Mineral Intelligence +1

2.china

The undisputed leader, accounting for **12.9 million sales** (over 60% of the global total). Nearly half of all new car sales in China are now electric.

- **Europe:** A recovering market with **4.3 million sales** in 2025, up 33% year-on-year.
- **North America:** Lagging behind with **1.8 million sales** in 2025, reflecting a 4% decline as some tax credits were modified or removed.
- **Emerging Markets:** Exceptional growth in Southeast Asia and Brazil, where sales surged by nearly **48% to 50%** in 2025.

3.Electric Bikes (Two-Wheelers)



Electric two-wheelers represent the most electrified vehicle segment globally, with a **15% sales share** in 2025.

Virta Global +1

- **High Volume:** In 2023 alone, **40 million e-bikes** were sold globally, often outpacing new car sales in total units.
- **Regional Strongholds:**

China

remains the dominant player, accounting for **73% of global sales**. However,

India

has emerged as the second-largest market for E2Ws.

- **Adoption Rates:**

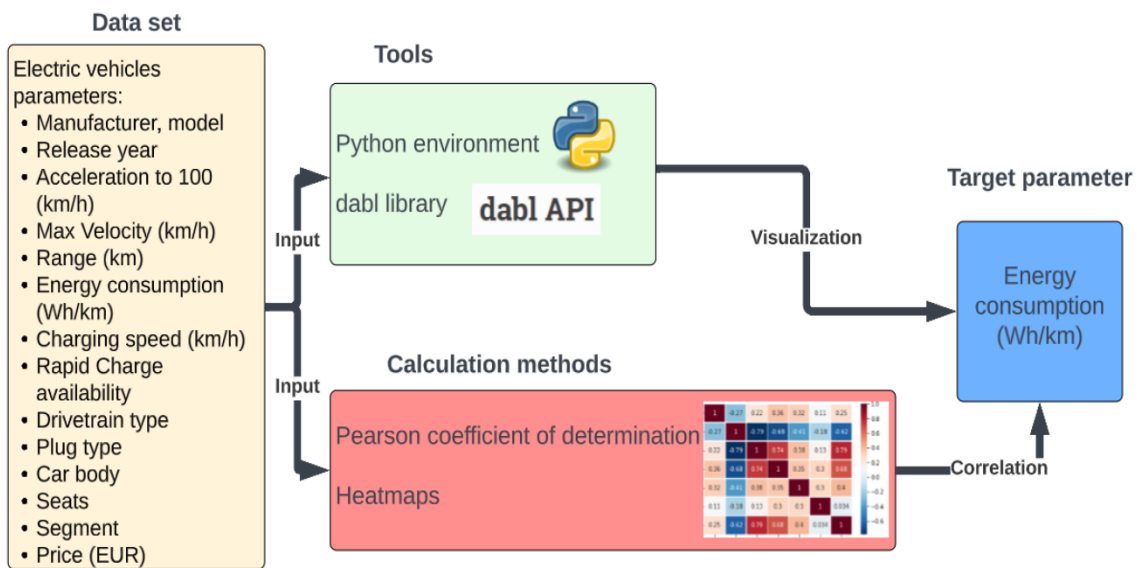
China

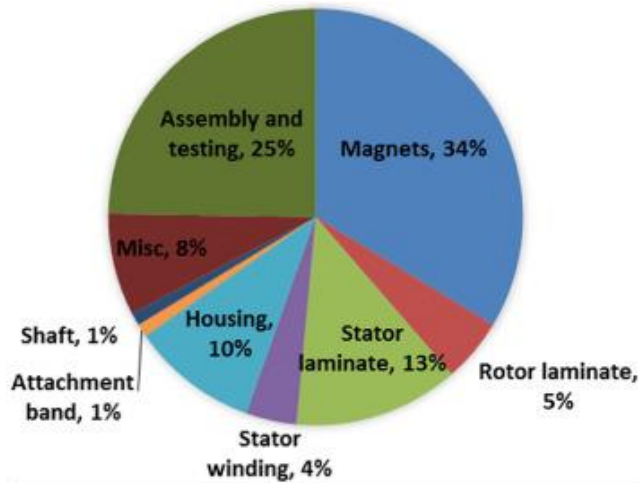
has the highest penetration at **54.8%**, followed by Vietnam at **21.7%**.

- **Market Value:** The electric motorcycle market was valued at **\$61 billion** last year and is projected to reach **\$109 billion** by 2030.

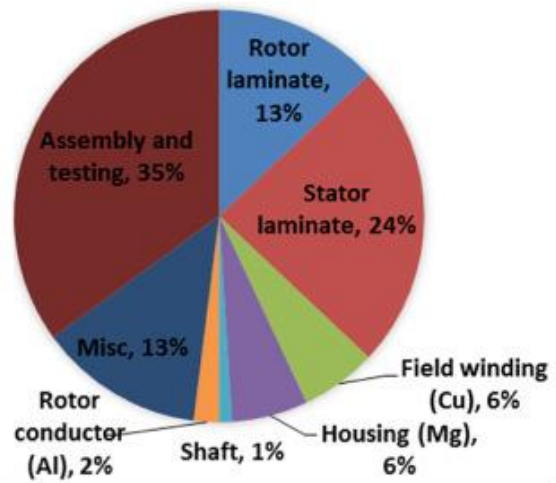
Key Differences in Regional Adoption

Feature	Car-Centric Regions (US/EU)	Bike-Centric Regions (Asia/Africa)
Primary EV Mode	Electric Passenger Cars (BEV/PHEV)	Electric Scooters & Motorcycles (E2W)
Main Driver	Emissions targets & premium tech	Affordability & urban congestion
Infrastructure	Large-scale public charging networks	Battery swapping & home charging
Cost Barrier	High; average price gap remains 20-30%	Low; many models already at price parity





(a) 32kW PM motor, \$522



(b) 40kW AI Induction motor, \$450

Essential tools for electric vehicles (EVs) in India range from everyday maintenance items like portable chargers and tire inflators to specialized insulated toolkits for high-voltage repairs. Because EVs operate on high-voltage systems (often over 500V), using standard metal tools can be dangerous; therefore, VDE-approved insulated tools are critical for any electrical work.



A smart home option that includes ARAI certification and "Auto Cut-Off" technology to protect battery health by preventing overcharging. It supports dynamic load balancing to adjust power based on your home's total electrical load.

Key Specifications and Requirements

When choosing a charger, verify your home's electrical capacity and your vehicle's port type.

- **Circuit Requirements:** A 40A charger generally requires a 50A breaker, while a 48A charger needs a 60A breaker.
- **Connectors:** Most modern EVs in Europe and India use **Type 2** connectors for AC charging. In North America, **J1772** (Type 1) and **NACS** (Tesla) are the standard.

Installation: **Level 2 chargers must typically be installed by a professional electrician to ensure proper grounding and compliance with local codes.**

Comparison of Popular Models

Charger Model	Power Output	Connector Type	Key Benefit
ChargePoint Home Flex	Up to 50A	J1772 or NACS	Most powerful and best overall app
Tesla Universal Wall Connector	48A	Both (Integrated)	Best for mixed-EV households
Emporia Level 2	48A	J1772 or NACS	Best budget and load management
Grizzl-E Classic	40A/48A	J1772 or NACS	Best for harsh/cold climates
Exicom Smart Spin Air	7.4kW–22kW	Type 2	Alexa-enabled with built-in 4G

EV chargers are categorized by their speed and power source, with three primary levels defining how quickly a car's battery is replenished

EV Charging Levels & Speeds

- **Level 1 (Slow AC Charging):** Uses a standard **120V household outlet** (common in North America) and typically comes with the car.
 - **Speed:** Adds roughly **3–5 miles (5–8 km)** of range per hour.
 - **Best For:** Overnight charging for drivers with low daily mileage.



- **Level 2 (Fast AC Charging):** Requires a dedicated **208V–240V circuit**, similar to large appliances like dryers.
 - **Speed:** Adds **12–40 miles (20–65 km)** of range per hour.
 - **Best For:** Daily home, workplace, or shopping center charging.

- **Level 3 / DC Fast Charging:** Bypasses the car's onboard converter to send **direct current (DC)** straight to the battery.
- **Speed:** Can charge a battery from **10% to 80% in 20–30 minutes**.
- **Best For:** Long-distance travel and highway rest stops.

2. Common Connector Types

Connector standards vary widely by region and charging speed:

- **J1772 (Type 1):** The standard for Level 1 and 2 AC charging in **North America and Japan**.
- **Mennekes (Type 2):** The standard for AC charging in **Europe and India**; it supports three-phase power for faster home charging.
- **CCS (Combined Charging System):** An enhanced version of Type 1 (CCS1) or Type 2 (CCS2) that adds two pins for DC fast charging. It is the most common fast-charging standard globally.
- **NACS (J3400):** Originally Tesla's proprietary plug, now becoming the standard in North America for many other brands (Ford, GM, etc.).

3. Home vs. Public Charging

Feature	Home Charging	Public Charging
Convenience	Highly convenient; charge while you sleep	Opportunistic; charge while working/shopping
Cost	Typically the cheapest option (residential rates)	Often 3–5 times more expensive than home rates
Installation	Requires an upfront cost for a wallbox	No installation needed for the user
Battery Health	Slower AC charging is generally better for long-term health	Frequent DC fast charging can cause faster degradation

11. Smart Charging Features

Modern chargers often include "smart" capabilities to optimize energy use:

- **Scheduling:** Set your car to charge during **off-peak hours** (e.g., midnight to 6 AM) when electricity rates are lowest.
- **Load Balancing:** Automatically adjusts the power drawn by the charger to prevent overloading your home's main electrical panel.
- **Solar Integration:** Some chargers, like the

Zappi

or **VOOL**, can prioritize using excess energy from your rooftop solar panels.



Analyzing EV range requires understanding the interplay between battery technology, environmental conditions, and testing standards. While "10 pages" of data is vast, the following comprehensive analysis covers the critical pillars that define EV range in 2025–2026.

Global Testing Standards: Why Ratings Vary

Manufacturers use different lab-based protocols, which explains why the same car can have multiple advertised ranges.

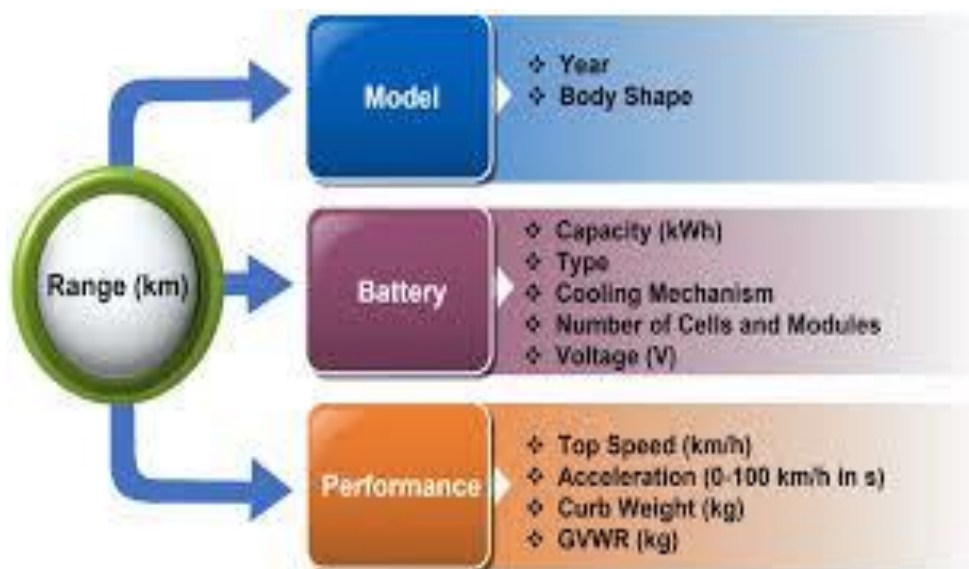
- **EPA (USA):** Generally the **strictest** and most accurate. It uses a correction factor (typically 0.7x) to account for real-world usage.

- **WLTP (Europe/India/Global):** More realistic than older tests but still roughly **10–15% more optimistic** than EPA.
- **CLTC (China):** The most **generous** standard, often showing figures **30% higher** than EPA because it prioritizes low-speed urban driving.
- **NEDC (Outdated):** Largely abandoned due to extreme inaccuracy, though still referenced in older data

2. Environmental & Technical Impact Factors

Real-world range often falls below rated figures due to external "range killers."

- **Temperature:** EVs perform optimally at **21.5°C (70°F)**.
 - **Cold:** Range can drop by **30–50%** at sub-zero temperatures due to battery chemistry and the massive energy draw of heating systems.
 - **Heat:** Range loss is less severe (roughly **15–20%**) but stems from battery cooling and air conditioning.
- **Speed & Aerodynamics:** Drag increases with the **square of speed**. For every **10 km/h** increase above highway speeds, energy consumption typically rises by **15%**.
- **Terrain:** Uphill driving drastically increases energy consumption, though **regenerative braking** can recover some energy on the descent.



Range analysis for Electric Vehicles (EVs)

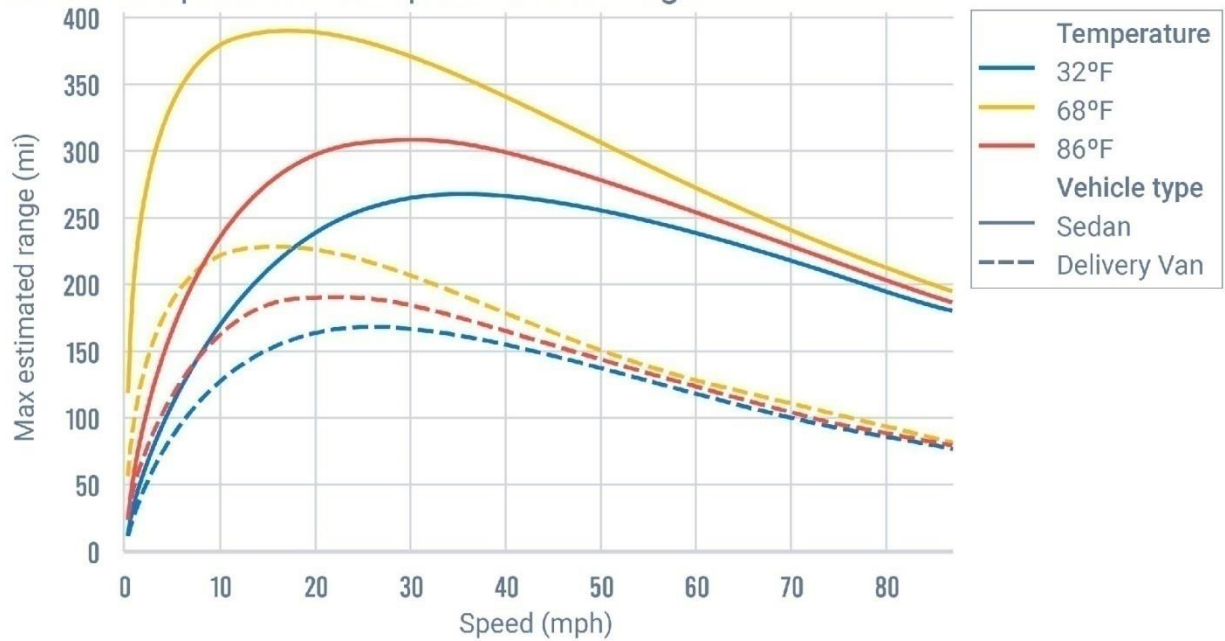
involves evaluating the distance a vehicle can travel on a single charge based on a complex interplay of technical, environmental, and behavioral variables. While manufacturers provide official ratings using standardized tests like WLTP or EPA, real-world performance often varies by **20–30%** due to dynamic conditions.

Key Determinants of EV Range

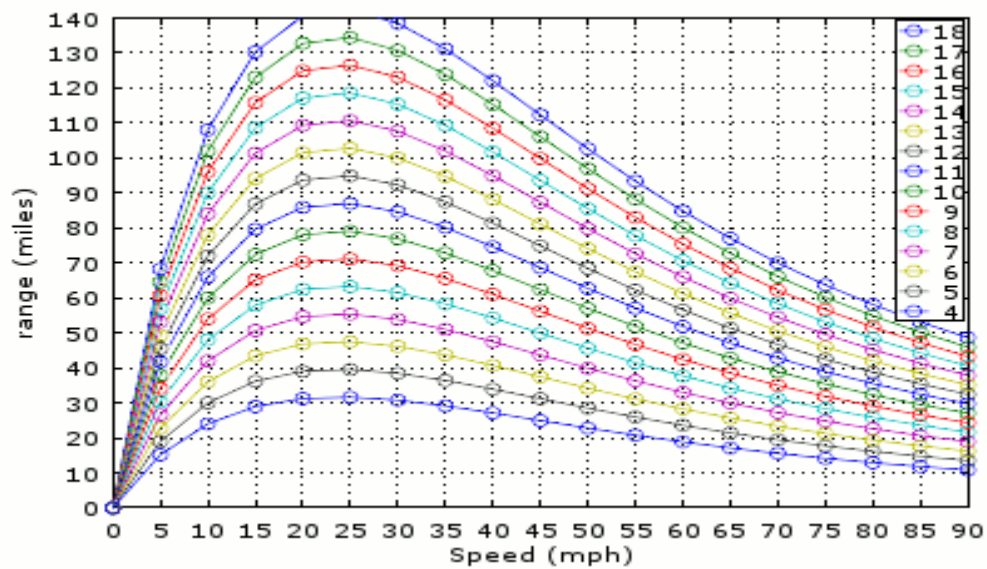
Range is influenced by both constant design parameters and variable operational factors:

- **Technical Factors:**
 - **Battery Capacity & Health:** Larger capacities (measured in kWh) provide a higher base range, but battery degradation over time typically reduces capacity by **2–3% per year**.
 - **Vehicle Design:** Aerodynamic drag (especially above 60 mph) and curb weight are critical. Heavier vehicles require more energy to move, with every **100 kg reduction** potentially increasing range by **10–11%**.
- **Environmental Factors:**
 - **Temperature:** EVs perform optimally at roughly **21.5°C (70°F)**. In extreme cold (–15°C), range can drop by up to **50%** due to reduced battery efficiency and high heater demand.
 - **Terrain:** Uphill driving significantly increases energy consumption, though [regenerative braking](#) on downhill stretches can recoup up to **20%** of energy.
- **User Behavior:**
 - **Driving Style:** Aggressive driving (rapid acceleration/hard braking) can reduce range by over **30%** compared to "Eco" modes or smooth driving.
 - **Speed:** Driving at 70 mph can consume **15–25% more energy** than at 60 mph due to exponential increases in air re

Effect of speed and temperature on range



2015/2016 SparkeV ranges for various battery kWh



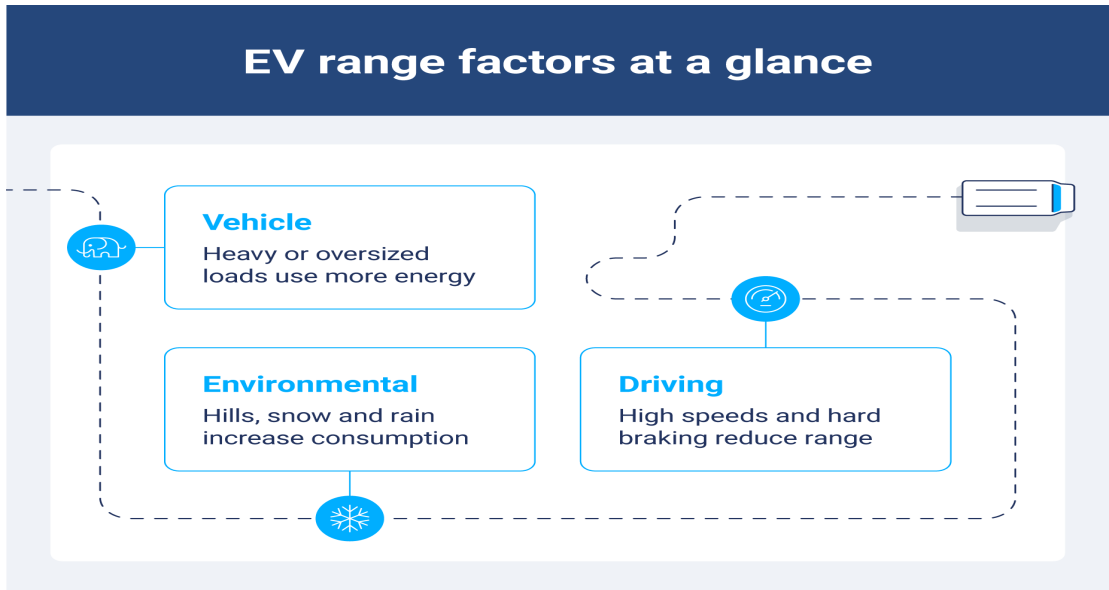
Methods of Range Prediction

Modern range analysis uses two primary methodologies to provide accurate "distance to empty" (DTE) estimates:

- 1. **History-Based Estimation:** Uses recent consumption data (e.g., the last 50 km) to project remaining range. While simple, it lacks accuracy during sudden changes in terrain or weather.
- 2. **Model-Based & Machine Learning:** Advanced models like [Deep Neural Networks \(DNN\)](#) and [Random Forest](#) incorporate real-time GPS data, elevation maps, and weather forecasts. These models can achieve prediction accuracy within **4–10 km** of actual remaining range.

Standardized Test Comparisons (2026 Context)

Test Cycle	Region	Characteristics	Realistic Accuracy
EPA	North America	Stringent; multi-cycle including highway/AC	Often viewed as most realistic.
WLTP	Europe/Global	Mixed urban/motorway; 23.25 km length	Typically 10–20% higher than real-world.
NEDC	Older Europe	Highly optimistic; mostly urban focus	Often 30% higher than real-world.



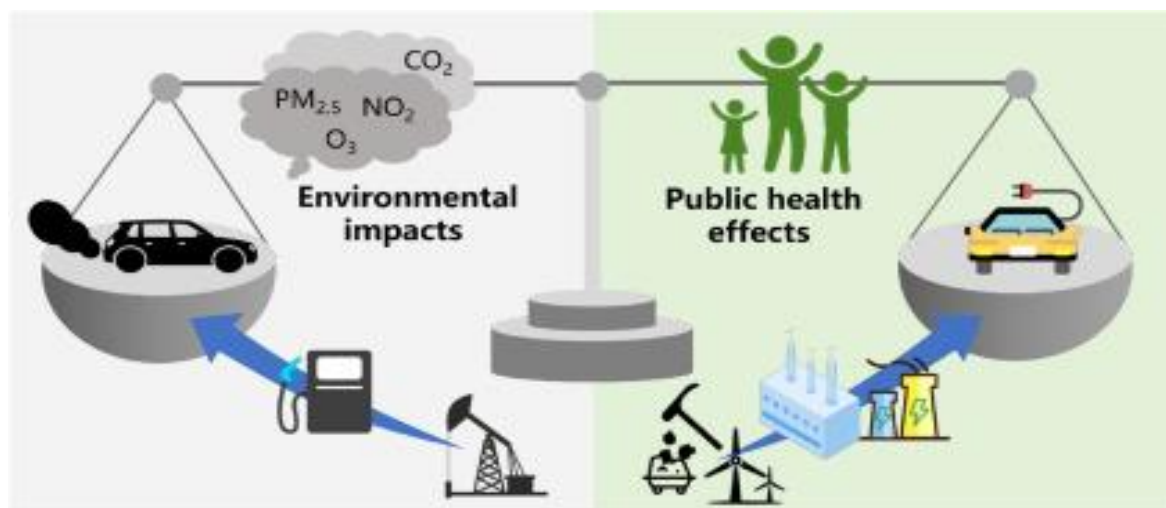
Electric vehicles (EVs), including cars and bikes, serve a wide range of practical purposes across personal, commercial, and environmental sectors. Their utility extends beyond simple transportation to innovative roles in energy management and urban sustainability.

1. Personal and Urban Commuting

- **Daily Commute:** EVs are ideal for short-to-medium distance travel, offering a smooth, quiet, and easy-to-drive experience without gears or complicated controls.
- **Traffic Navigation:** Electric bikes and scooters are [nimble and efficient](#), allowing riders to bypass heavy urban congestion and navigate narrow streets where cars might struggle.
- **First and Last-Mile Connectivity:** Shared electric mobility services (like rental e-bikes) bridge the gap between homes and public transport hubs, making urban travel more streamlined.
- **Convenience:** Many EVs can be [charged at home](#) using standard outlets, eliminating the need to visit crowded fuel stations

4. Environmental and Public Health

- **Emission Reduction:** EVs produce [zero tailpipe emissions](#), directly improving local air quality and helping cities meet climate targets.
- **Noise Reduction:** Their silent operation significantly [reduces noise pollution](#) in busy residential and urban areas.
- **Sustainable Tourism:** E-bikes allow tourists to [explore remote areas and trails](#) without the physical strain of traditional cycling or the environmental impact of gas-powered vehicles.



When it comes to the "**Green and Clean**" side of EVs, the benefits go way beyond just avoiding the petrol pump. Switching to electric cars and bikes acts as a major upgrade for both the planet and our personal well-being.

1. Breathing Easier: Local Air Quality

The most immediate win is the elimination of **tailpipe emissions**.

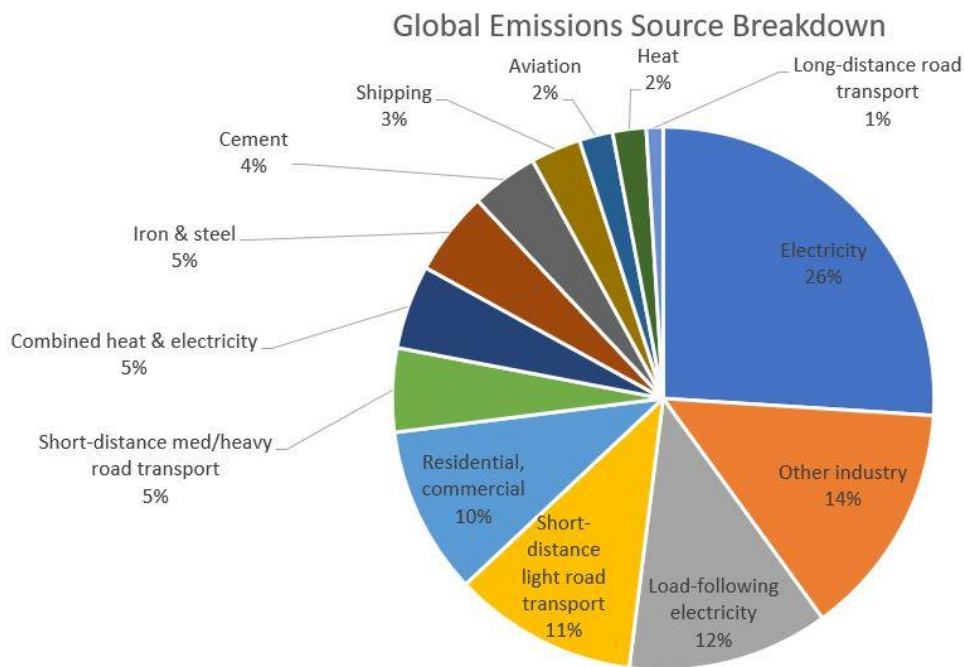
- **Pollutant Reduction:** Unlike ICE vehicles, EVs don't spit out Nitrogen Oxides (NO_x), Carbon Monoxide, or particulate matter (PM_{2.5}) into the streets. This is a game-changer for improving urban air quality.
- **Health Impact:** Cleaner air directly correlates to lower rates of respiratory issues, asthma, and cardiovascular diseases in densely populated areas.



3. Climate Action: Carbon Footprint

Even when considering the manufacturing process, EVs are significantly more sustainable over their lifecycle.

- **Net-Zero Goals:** By using electricity (especially when sourced from [solar or wind](#)), EVs help slash the greenhouse gas emissions that drive global warming.
- **Second Life:** EV batteries aren't just "thrown away." They are increasingly being repurposed as [energy storage systems](#) for homes and the grid before being recycled.



A carbon footprint measures the total greenhouse gases, primarily, emitted by human activities (individuals, products, or organizations), usually expressed in tonnes of -equivalent (tCO₂e). Climate action involves reducing this footprint through energy efficiency, low-carbon transport, diet changes, and renewable energy to mitigate global warming.

Clean Data from Excel, CSV, PDF, and Google Sheets with Data Interpreter

Applies to: Tableau Cloud, Tableau Desktop, Tableau Server

When you track data in Excel spreadsheets, you create them with the human interface in mind. To make your spreadsheets easy to read, you might include things like titles, stacked headers, notes, maybe empty rows and columns to add white space, and you probably have multiple tabs of data too.

When you want to analyze this data in Tableau, these aesthetically pleasing attributes make it very difficult for Tableau to interpret your data. That's where Data Interpreter can help.

Tip: Though Tableau's Excel add-in is no longer supported, Data Interpreter can help you reshape your data for analysis in Tableau.

What does Data Interpreter do?

Data Interpreter can give you a head start when cleaning your data. It can detect things like titles, notes, footers, empty cells, and so on and bypass them to identify the actual fields and values in your data set.

It can even detect additional tables and sub-tables so that you can work with a subset of your data independently of the other data.

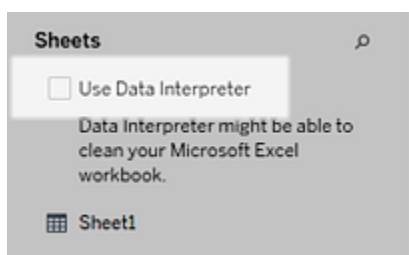
After Data Interpreter has done its magic, you can check its work to make sure it captured the data that you wanted and identified it correctly. Then, you can make any necessary adjustments.

After you select the data that you want to work with, you might also need to do some additional cleaning steps like pivoting your data, splitting fields, or adding filters to get the data in the shape you want before starting your analysis.

Note: If your data needs more cleaning than what Data Interpreter can help you with, try [Tableau Prep](#)([Link opens in a new window](#)).

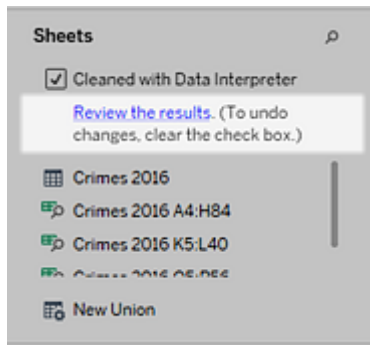
Turn on Data Interpreter and review results

1. From the **Connect** pane, connect to an Excel spreadsheet or other connector that supports Data Interpreter such as Text (.csv) files, PDF files or Google sheets.
2. Drag a table to the canvas (if needed), then on the **Data Source** page, in the left pane, select the **Use Data Interpreter** check box to see if Data Interpreter can help clean up your data.

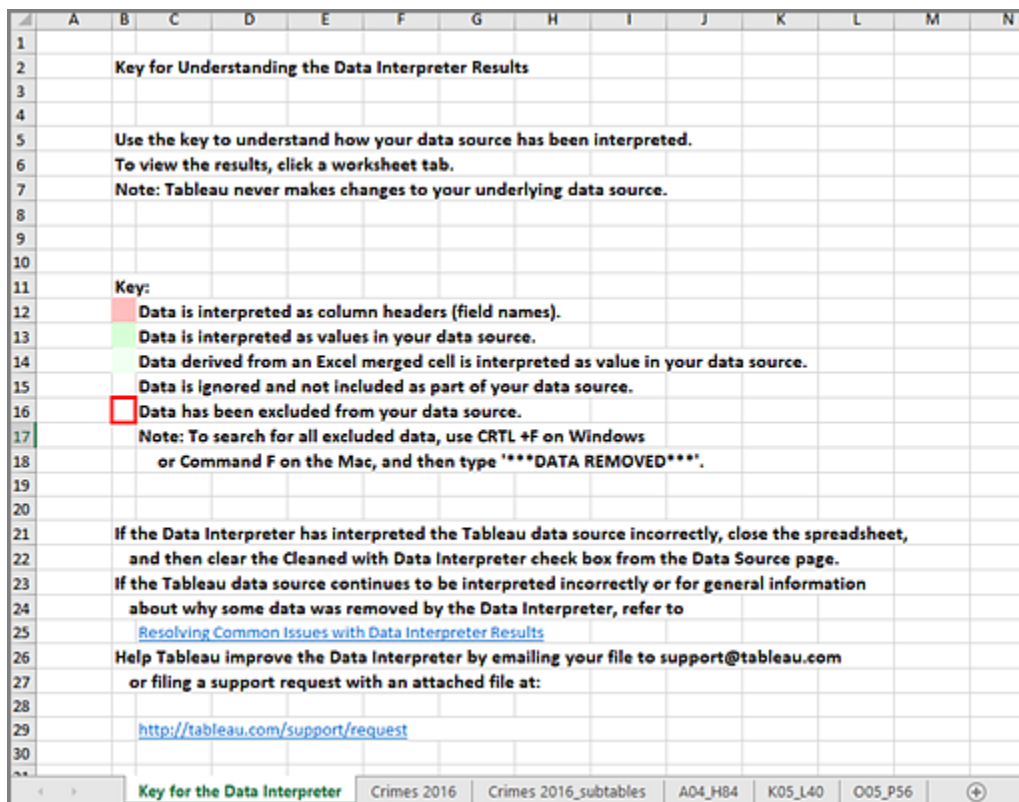


Note: When you clean your data with Data Interpreter, Data Interpreter cleans all the data associated with a connection in the data source. Data Interpreter does not change the underlying data.

3. In the Data pane, click the **Review the results** link to review the results of the Data Interpreter.



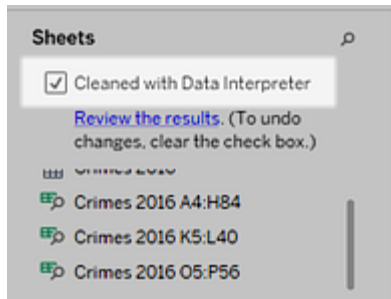
A copy of your data source opens in Excel on the **Key for the Data Interpreter** tab. Review the key to find out how to read the results.



4. Click each tab to review how Data Interpreter interpreted the data source.

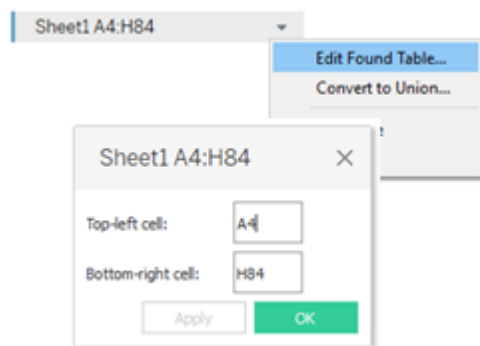
If Data Interpreter found additional tables, also called found tables or sub-tables, they are identified in the <sheet name>_subtables tab by outlining their cell ranges. A separate tab is also included for each sub-table, color coded to identify the header and data rows.

If Data Interpreter does not provide the expected results, clear the **Cleaned with Data Interpreter** check box to use the original data source.



5. To replace the current table with any of the found tables, drag the current table off the canvas and then drag the found table that you want to use to the canvas.

If Data interpreter has misidentified the range of the found table, after you drag the found table to the canvas, click the drop-down arrow on that table, and then select **Edit Found Table** to adjust the corners of the found table (the top-left cell and bottom-right cell of the table).



6. After you have the data that you want to work with, you can apply any additional cleaning operations to your data so that you can analyze it.

Data Interpreter Example

In this example we are connecting to an Excel spreadsheet with violent crime data by city and state for the year 2016. This spreadsheet includes multiple tables on one sheet and some extra formatting.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Violent Crimes in 2016 in the United States by City and State																
2																	
3																	
4	Location		Months						state		Total Crimes 2016		State		Population 2016		
5	city	state	Apr	Jun	Jul	Aug	Sep	Oct									
6	Albuquerque	New Mexico						46			12	Alabama	4860545		Alabama	4860545	
7	Anaheim	California			4						26	Alaska	741522		Alaska	741522	
8	Anchorage	Alaska		1				26			132	Arizona	6908642		Arizona	6908642	
9	Arlington	Texas					17				515	California	2988231		Arkansas	2988231	
10	Atlanta	Georgia						85			64	Colorado	39296476		California	39296476	
11	Aurora	Colorado						16			105	D.C.	5530105		Colorado	5530105	
12	Austin	Texas					28				210	Florida	3587685		Connecticut	3587685	
13	Bakersfield	California			22						85	Georgia	952698		Delaware	952698	
14	Baltimore	Maryland								230	9	Hawaii	684336		District of Co	684336	
15	Boston	Massachusetts						28			536	Illinois	20656589		Florida	20656589	
16	Buffalo	New York						38			151	Indiana	10313620		Georgia	10313620	
17	Chandler	Arizona						3			10	Kansas	1428683		Hawaii	1428683	
18	Charlotte-M	North Carolina			25						99	Kentucky	1680026		Idaho	1680026	
19	Chicago	Illinois									127	Louisiana	12835726		Illinois	12835726	
20	Chula Vista	California		2			1				230	Maryland	6634007		Indiana	6634007	
21	Cincinnati	Ohio						50			28	Massachuset	3130869		Iowa	3130869	
22	Cleveland	Ohio						89			221	Michigan	2907731		Kansas	2907731	
23	Colorado Sp	Colorado					15				26	Minnesota	4436113		Kentucky	4436113	
24	Columbus	Ohio						70			223	Missouri	4686157		Louisiana	4686157	
25	Corpus Chris	Texas			9						29	Nebraska	1330232		Maine	1330232	
26	Dallas	Texas					118				128	Nevada	6024752		Maryland	6024752	
27	Denver	Colorado					33				86	New Jersey	6823721		Massachuset	6823721	
28	Detroit	Michigan		5				221			46	New Mexico	9933445		Michigan	9933445	
29	Durham	North Carolina							30		290	New York	5525050		Minnesota	5525050	
30	El Paso	Texas						14			82	North Carolin	2985415		Mississippi	2985415	
31	Fort Wayne	Indiana						34			217	Ohio	6091176		Missouri	6091176	
32	Fort Worth	Texas		7				49			82	Oklahoma	1038656		Montana	1038656	
33	Fresno	California				19					14	Oregon	1907603		Nebraska	1907603	
34	Greensboro	North Carolina							20		259	Pennsylvania	2939254		Nevada	2939254	

- A. Title
- B. Merged header cells
- C. Extra white space
- D. Sub-tables

The extra formatting in this spreadsheet makes it difficult for Tableau to determine what the field headers and values are.

Instead, it reads the data vertically and assigns each column the default value F1, F2, F3 (Field 1, Field 2, Field 3) and so on. Blank cells are read as null values.

Crimes 2016 (crimes_2016)

Connection: ☒ Live ☐ Extract

Filters: 0 | Add

Crimes 2016

Sort fields: Data source order

Show aliases Show hidden fields 82 rows

crimes_2016 F1	crimes_2016 F2	crimes_2016 F3	crimes_2016 F4	crimes_2016 F5	crimes_2016 F6	crimes_2016 F7	crimes_2016 F8	crimes_2016 F9	crimes_2016 F10	crimes_2016 F11	crimes_2016 F12	crimes_2016 F13
Violent Crimes in 201...	null	null	null	null	null	null	null	null	null	null	null	null
Location	null	Month	null	null	null	null	null	null	null	null	null	null
city	state	Apr	Jun	Jul	Aug	Sep	Oct	state	Total Crimes 2016	State	Population 2016	
Albuquerque	New Mexico	null	null	null	null	46	null	Alabama	12	Alabama	4860545	
Anaheim	California	null	4	null	null	null	null	Alaska	26	Alaska	741522	
Anchorage	Alaska	1	null	null	null	26	null	Arizona	132	Arizona	6909642	
Arlington	Texas	null	null	17	null	null	null	California	515	Arkansas	2988231	
Atlanta	Georgia	null	null	null	null	85	null	Colorado	64	California	39296476	

To see if Data Interpreter can help clean this data set, we select **Use Data Interpreter**.

Data Interpreter detected the proper headings for the fields, removed the extra formatting and found several sub-tables. The sub-tables are listed in the **Sheets** section in the Data pane and are named using the original sheet name and the cell ranges for each sub-table.

In this example there are three sub-tables: **Crimes 2016 A4:H84**, **Crimes 2016 K5:L40**, and **Crimes 2016 O5:P56**.

Crimes 2016 (crimes_2016) (2)

Connection: ☒ Live ☐ Extract

Filters: 0 | Add

Crimes 2016

Sort fields: Data source order

Show aliases Show hidden fields 79 rows

crimes_2016 Location city	crimes_2016 Location state	crimes_2016 Months Apr	crimes_2016 Months Jun	crimes_2016 Months Jul	crimes_2016 Months Aug	crimes_2016 Months Sep	crimes_2016 Months Oct	crimes_2016 state	crimes_2016 Total Crimes 2016	crimes_2016 State	crimes_2016 Population 2016
Albuquerque	New Mexico	null	null	null	null	46	null	Alabama	12	Alabama	4,860,545
Anaheim	California	null	4	null	null	null	null	Alaska	26	Alaska	741,522
Anchorage	Alaska	1	null	null	null	26	null	Arizona	132	Arizona	6,909,642
Arlington	Texas	null	null	null	17	null	null	California	515	Arkansas	2,988,231
Atlanta	Georgia	null	null	null	null	85	null	Colorado	64	California	39,296,476
Aurora	Colorado	null	null	null	null	16	null	D.C.	106	Colorado	6,530,106
Austin	Texas	null	null	null	28	null	null	Florida	210	Connecticut	3,587,685
Bakersfield	California	null	22	null	null	null	null	Georgia	86	Delaware	952,698
Baltimore	Maryland	null	null	null	null	null	230	Massachusetts	9	Department of Columbia	684,336

To examine the results of the Data Interpreter more closely, we click the **Review the results** link in the Data pane to view an annotated copy of the spreadsheet.

Here we see a copy of the original data, color coded to identify which data was identified as header data and which data was identified as field values.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Violent Crimes in 2016 In the United States by City and State																
2																	
3																	
4	Location	Location	Months	Months	Months	Months	Months	Months									Header
5	city	state	Apr	Jun	Jul	Aug	Sep	Oct			state	Total Crimes 2016			State	Populatio	Header
6	Albuquerque	New Mexico						46			Alabama	12			Alabama	4860545	Data
7	Anaheim	California			4						Alaska	26			Alaska	741522	Data
8	Anchorage	Alaska	1					26			Arizona	132			Arizona	6908642	Data
9	Arlington	Texas					17				California	515			Arkansas	2988231	Data
10	Atlanta	Georgia						85			Colorado	64			California	39296476	Data
11	Aurora	Colorado						16			D.C.	105			Colorado	5530105	Data
12	Austin	Texas					28				Florida	210			Connectic	3587685	Data
13	Bakersfiel	California			22						Georgia	85			Delaware	952698	Data
14	Baltimore	Maryland							230		Hawaii	9			District of	684336	Data
15	Boston	Massachusetts						28			Illinois	536			Florida	20656589	Data
16	Buffalo	New York						38			Indiana	151			Georgia	10313620	Data
17	Chandler	Arizona						3			Kansas	10			Hawaii	1428683	Data

The next tab shows us the sub-tables that Data Interpreter found, outlined by the cell ranges.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Violent Crimes in 2016 In the United States by City and State																
2																	
3																	
4	Location	Location	Months	Months	Months	Months	Months	Months									
5	city	state	Apr	Jun	Jul	Aug	Sep	Oct			state	Total Crimes 2016			State	Population 2016	
6	Albuquerque	New Mexico						46			Alabama	12			Alabama	4860545	
7	Anaheim	California			4						Alaska	26			Alaska	741522	
8	Anchorage	Alaska	1					26			Arizona	132			Arizona	6908642	
9	Arlington	Texas					17				California	515			Arkansas	2988231	
10	Atlanta	Georgia						85			Colorado	64			California	39296476	
11	Aurora	Colorado						16			D.C.	105			Colorado	5530105	
12	Austin	Texas					28				Florida	210			Connectic	3587685	
13	Bakersfiel	California			22						Georgia	85			Delaware	952698	
14	Baltimore	Maryland							230		Hawaii	9			District of	684336	
15	Boston	Massachusetts						28			Illinois	536			Florida	20656589	
16	Buffalo	New York						38			Indiana	151			Georgia	10313620	
17	Chandler	Arizona						3			Kansas	10			Hawaii	1428683	
18	Charlotte	North Carolina			25						Kentucky	95			Idaho	1680026	
19	Chicago	Illinois							536		Louisiana	127			Illinois	12835726	
20	Chula Vist	California	2				1				Maryland	230			Indiana	6634007	
21	Cincinnati	Ohio						50			Massachu	28			Iowa	3130869	
22	Cleveland	Ohio						89			Michigan	221			Kansas	2907731	
23	Colorado	Colorado					15				Minnesot	26			Kentucky	4436113	
24	Columbus	Ohio						70			Missouri	223			Louisiana	4686157	
25	Corpus Ch	Texas			9						Nebraska	29			Maine	1330232	
26	Dallas	Texas					118				Nevada	128			Maryland	6024752	
27	Denver	Colorado					33				New Jerse	86			Massachu	6823721	
28	Detroit	Michigan	5					221			New Mexi	46			Michigan	9933445	
29	Durham	North Carolina							30		New York	290			Minnesoti	5525050	
30	El Paso	Texas						14			North Car	82			Mississipp	2985415	
31	Fort Wayr	Indiana						34			Ohio	217			Missouri	6091176	

In this example the first sub-table, **Crimes 2016 A4:H84**, has the main data that we want to work with. To use this table as our data table, we can simply drag the original table off the canvas and then drag the new table to the canvas.

←

→

Connections

crimes_2016

Microsoft Excel

Crimes 2016

Crimes 2016 A4:H84

Crimes 2016 K5:L40

...

New Union

Crimes 2016 A4:H84 (crimes_2016)

Connection

Live

Extract

Filters

0

Add

Crimes 2016 A4:H84

✓

Cleaned with Data Interpreter

Review the results. (To undo changes, clear the check box.)

Crimes 2016

Crimes 2016 A4:H84

Crimes 2016 K5:L40

...

New Union

Sort fields

Data source order

Show aliases

Show hidden fields

79

rows

Crimes 2016 A4:H84	Crimes 2016 A4:H84	Crimes 2016 A4:H84	Crimes 2016 A4:H84	Crimes 2016 A4:H84	Crimes 2016 A4:H84	Crimes 2016 A4:H84	Crimes 2016 A4:H84
Location city	Location state	Months Apr	Months Jun	Months Jul	Months Aug	Months Sep	Months Oct
Albuquerque	New Mexico	null	null	null	null	46	null
Anaheim	California	null	4	null	null	null	null
Anchorage	Alaska	1	null	null	null	26	null
Arlington	Texas	null	null	null	17	null	null
Atlanta	Georgia	null	null	null	null	85	null
Aurora	Colorado	null	null	null	null	16	null
Austin	Texas	null	null	null	28	null	null
Bakersfield	California	null	22	null	null	null	null
Baltimore	Maryland	null	null	null	null	null	230
Boston	Massachusetts	null	null	null	null	28	null
Buffalo	New York	null	null	null	null	38	null
Chandler	Arizona	null	null	null	null	3	null

Once we have the data that we want to work with in the canvas, we can do some additional clean up on the data. For example we can:

- Change the field names so that they represent city, state, and month names.
- Pivot the months fields.
- Drag in the third sub-table **Crimes 2016 o5:P56** and join it to our first sub-table on the **State** field to include state populations for our analysis.
- Hide any duplicate fields that were added as a result of the join.

The results might look something like this:

Sort fields Data source order				
Crimes 2016 A4:H84 City	Crimes 2016 A4:H84 State	Crimes 2016 O5:P56 Population 2016	Abe Pivot Months	# Pivot Crimes
Phoenix	Arizona	6,908,642	August	111
Pittsburgh	Pennsylvania	12,787,085	August	null
Plano	Texas	27,904,862	August	5
Portland	Oregon	4,085,989	August	null
Raleigh	North Carolina	10,156,689	August	null
Riverside	California	39,296,476	August	7
Sacramento	California	39,296,476	August	null
San Antonio	Texas	27,904,862	August	null
San Diego	California	39,296,476	August	30
San Francisco	California	39,296,476	August	null
San Jose	California	39,296,476	August	35
Santa Ana	California	39,296,476	August	null
Seattle	Washington	7,280,934	August	14
St. Louis	Missouri	6,091,176	August	133
St. Petersburg	Florida	20,656,589	August	14

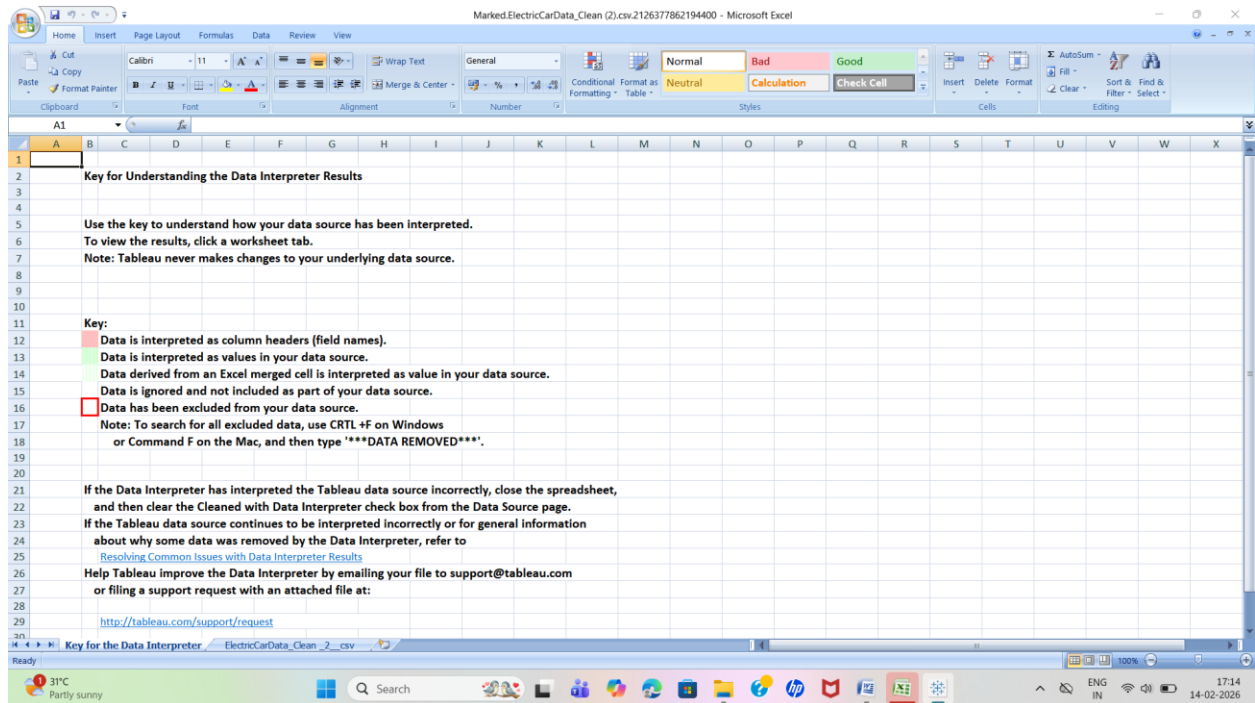
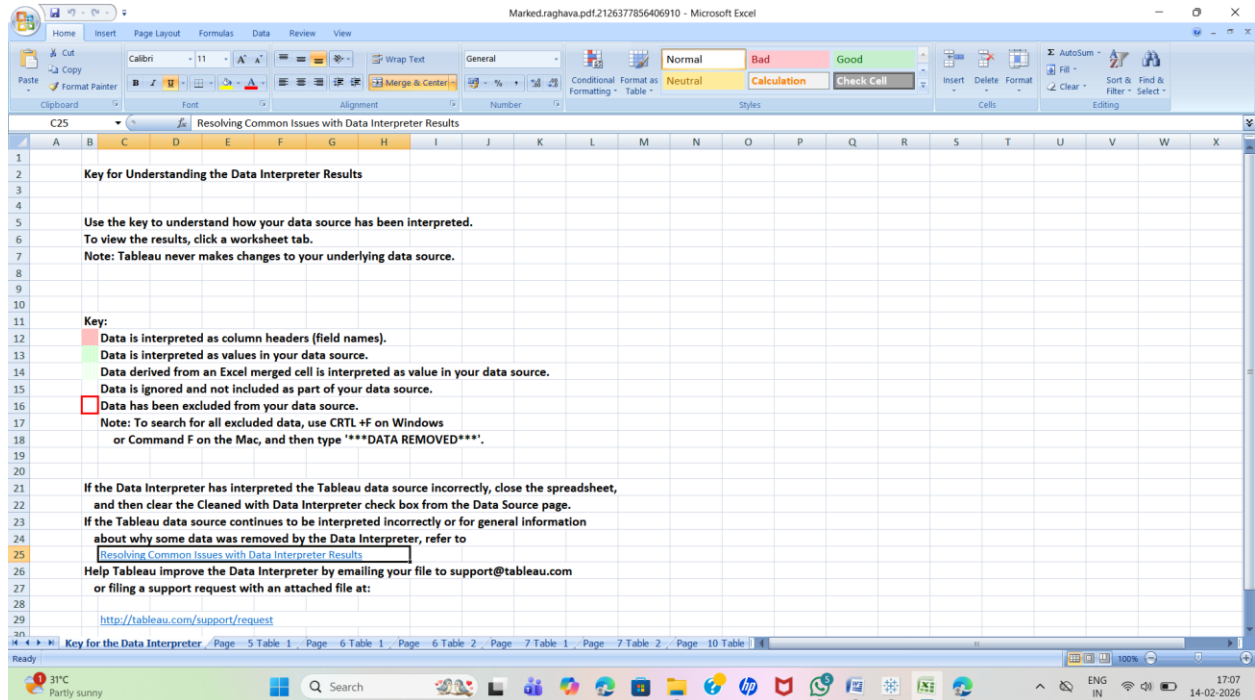
Now we are ready to start analyzing our data in Tableau.

When Data Interpreter is not available

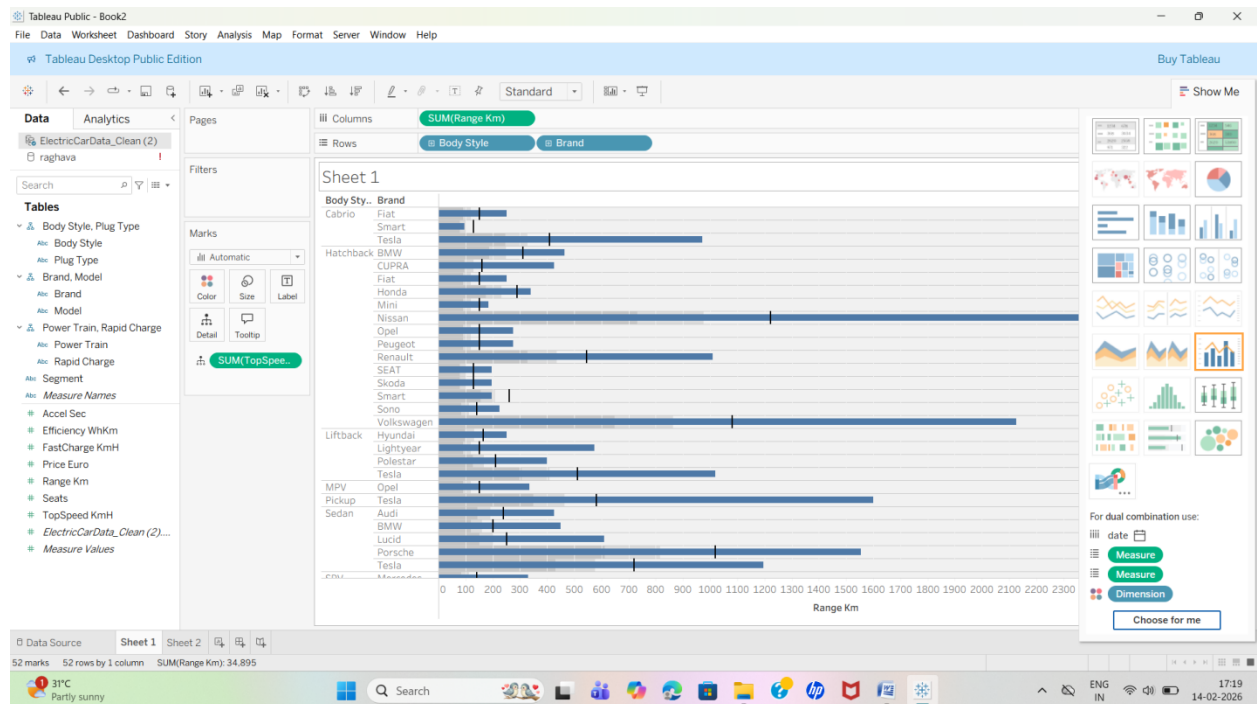
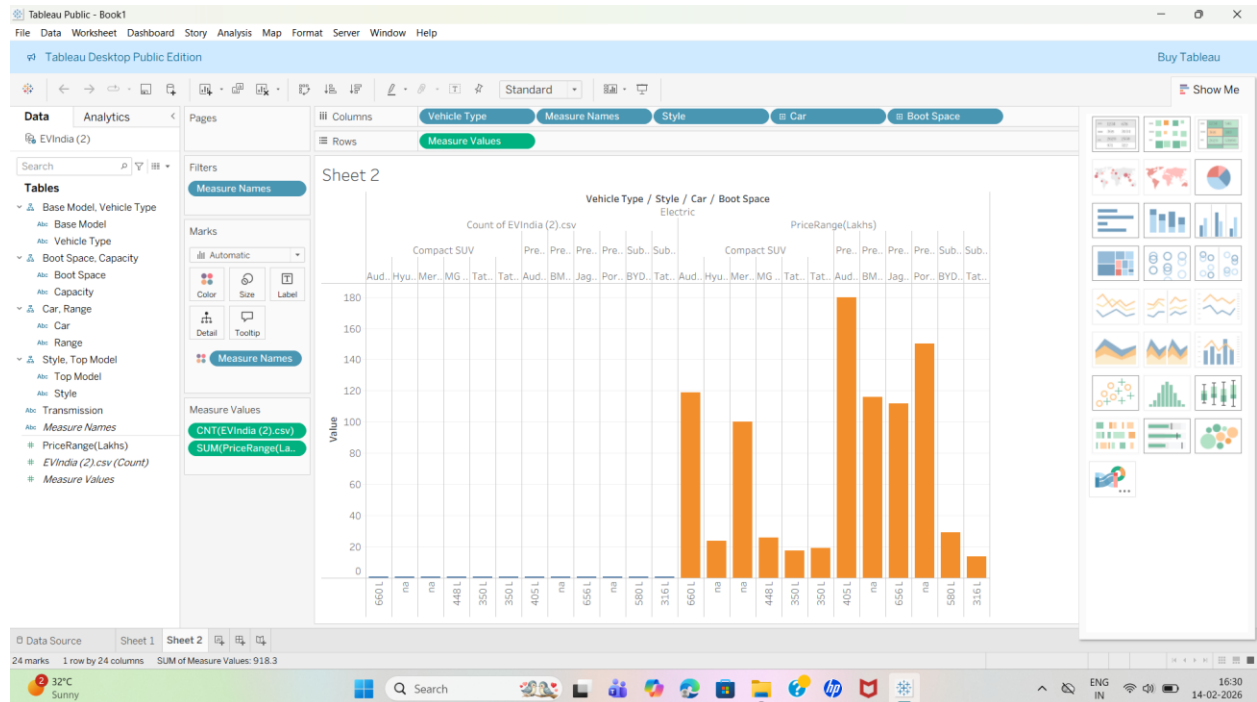
The Data Interpreter option might not be available for the following reasons:

- **The data source is already in a format that Tableau can interpret:** If Tableau Desktop doesn't need extra help from Data Interpreter to handle unique formatting or extraneous information, the Data Interpreter option is not available.
- **Many rows or many columns:** The Data Interpreter option is not be available when your data has the following attributes:
 - Data contains more than 2000 columns.
 - Data contains more than 3000 rows and more than 150 columns.
- **The data source is not supported:** Data Interpreter is only available for Microsoft Excel, Text (.csv) files, PDF files and Google Sheets. For Excel, your data must be in the .xls or .xlsx format.

RESULT



OUTPUTS



GROUP PHOTO



DEMO LINK :

https://drive.google.com/file/d/139OM1IJV7iurWik4IDVFhoiVr_CbXlbs/view?usp=drivesdk